PAPER TREE EDUCATIONAL STUDIO · JEE MAIN MATHEMATICS

# Errata report: v1.0.0 to v1.0.1

8 October 2026. This report covers the full errata round run on the released v1.0.0 bank (2,700 questions, 14 units) and the automated test suite run on the v1.0.1 release folder.

**2,699 / 2,700**
keys confirmed by independent code recomputation

**299**
audit findings (2 critical, 65 major, 232 minor)

**304**
closures, including 5 raised after the fixes

**122**
items with stem or options changed

**132 / 132**
changed items re-solved independently and agreed

**0**
key letters moved; unchanged items keep their exact option order

## 1. What was done

1. **Independent recomputation of every item.** Fresh verifier agents received only the stem and the four options of each released question, in packs of about 50, with no keys, drafts or checking code. For each question they wrote and ran their own sympy or brute-force code (2,679 of 2,700 by code; the rest are conceptual and were reasoned), evaluated all four options, and reported every option equal to their result. They agreed with 2,699 keys. The single disagreement (JEE-MATH-U10-140) was a verifier error, confirmed by the editor with sympy. All 342 integer answers agreed.
2. **Hostile unit audit.** Separate auditor agents read every full record (stem, options, key, explanation, distractor rationales, metadata) unit by unit and compared every pair within each slice. Categories: wrong key, ambiguity, explanation errors, integer alternates, distractor defects, duplicates and answer leaks, metadata, scope, presentation.
3. **Fixes.** Correcting agents re-verified each finding with sympy before acting. Rules: a critical finding leaves exactly one defensible option; a duplicate that reuses the same data or answer is rewritten, a merely related item is linked by an alias dependency so the paper generator never pairs the two; class labels follow NCERT placement; quotas and IDs never change. Every batch was re-run through the authoring harness (machine check, KaTeX render, house style) with 0 blocking results.
4. **Bank-wide edits.** Natural logarithms are now written log_e everywhere (87 items changed in notation only). 36 items moved to Class XI. 110 alias groups were added. Two integer alternates left the Section B pool (one now has a negative answer, one item lost its alternate in a rewrite) and were replaced like for like.
5. **Re-verification.** All 132 items whose stem, options or integer alternate changed were rebuilt and sent to fresh verifiers. That pass flagged the phrase "(possibly equal)" in two Unit 2 stems as ambiguous; the editor reworded those and a third stem using the same phrase, and all three were verified again. A post-fix similarity scan of every rewritten item against the whole bank was reviewed by a separate agent: 57 pairs independent, 1 linked by a new alias.
6. **Automated test suite.** 25 tests run against the finished release folder (section 6).

## 2. Findings by category

| Code | Category | Findings | Fixed | Rewritten | Aliased | Retained |
|---|---|---|---|---|---|---|
| E2 | Ambiguous or under-specified stem | 3 | 3 | 0 | 0 | 0 |
| E3 | Explanation error | 7 | 6 | 1 | 0 | 0 |
| E5 | Distractor defect | 37 | 37 | 0 | 0 | 0 |
| E6 | Duplicate or answer leak | 171 | 8 | 69 | 93 | 1 |
| E7 | Metadata (class, type, difficulty) | 34 | 33 | 1 | 0 | 0 |
| E8 | Scope | 9 | 2 | 2 | 0 | 5 |
| E9 | Presentation and notation | 38 | 36 | 0 | 0 | 2 |

Most findings were duplicates or answer leaks between related items (E6): a pair a paper could include together, where one item's stem or key hands over the other's answer. The 65 major E6 findings were rewritten or relinked; none affected the correctness of a key.

## 3. Critical findings

Two items were upgraded to critical because a second option could be defended. Both are fixed and re-verified. A third critical finding came from the re-verification pass.

**ERR-054 · U03-C-044 (JEE-MATH-U03-154) · Ambiguous or under-specified stem**

The stem adds 'non-symmetric' to rule out coincidences among the options, but that does not rule them out. For a non-symmetric orthogonal P (which is non-singular and satisfies PP^T = P^TP), options A (Q^2), B (I) and D (PP^T) all equal QQ^T. B is the only option that holds for every admissible P, but 'Then QQ^T equals' does not say 'for every such P', so a student can defend A or D.

*Closure (fixed):* Confirmed with sympy: for P = [[0,1,0],[-1,0,0],[0,0,1]] (non-symmetric, orthogonal, normal) Q^2 = PP^T = QQ^T = I. Stem now says 'real' and 'for every such P, QQ^T necessarily equals'; distractors PP^T and Q^2 replaced by -I and Q^T. Options now I, -I, Q, Q^T: Q = I iff P symmetric (excluded), so Q != I and Q^T = Q^{-1} != I; QQ^T has non-negative diagonal so never -I. Exactly one option holds for every admissible P. Rationales, values and explanation updated.

**ERR-079 · U05-B-030 (JEE-MATH-U05-077) · Ambiguous or under-specified stem**

Stem says 'For a positive integer n'; at n = 1 distractor C ($2n$) also equals the true ratio $2$, so C is a correct value for one admissible n.

*Closure (fixed):* Confirmed with sympy: at n = 1 distractor 2n equals the key 2. Stem now reads 'For every positive integer n'; distractor 2n replaced by (2n+1)/(n+1) (indices one more than given, mirroring the existing one-less slip (2n-1)/n). sympy: no option equals 2 and no two options coincide for any positive integer n; check extended to n = 1..10; rationales now show the computation.

**ERR-300 · U02-C-042 (JEE-MATH-U02-152) · Ambiguous or under-specified stem**

Re-verification: 'real and less than 5 (possibly equal)' can be read as 'less than or equal to 5'; under that reading k = 5 also qualifies and no option is correct.

*Closure (fixed):* Stem now reads '(not necessarily distinct) are real and strictly less than 5'. Key unchanged. Re-verified independently (V101-7).

## 4. Retained findings

- **ERR-142 U08-A-002** (E9): Log-notation finding; per errata instructions it is ignored here because the editor unifies \log notation bank-wide afterwards. No change to U08-A-002, 012, 017, 042, 050.
- **ERR-156 U08-C-029** (E9): Log-notation finding; per the errata instructions the editor unifies \log notation bank-wide afterwards. No change in this batch.
- **ERR-198 U10-D-010** (E6): Overlap was only with 163 (single substitution of a point into y^2 = 4ax). 163 (U10-D-013) has been rewritten into a focus-on-a-line task with the axis along the y-axis, so U10-D-010 no longer has a near-duplicate; no alias needed.
- **ERR-215 U10-F-025** (E8): Errata ruling: translated (shifted) standard conics are in scope. Completing squares only; no out-of-unit tools in the explanation.
- **ERR-216 U10-F-026** (E8): Errata ruling: translated (shifted) standard conics are in scope. Completing squares only; no out-of-unit tools in the explanation.
- **ERR-217 U10-F-034** (E8): Errata ruling: translated (shifted) standard conics are in scope. Completing squares only; no out-of-unit tools in the explanation.
- **ERR-218 U10-F-046** (E8): Errata ruling: translated (shifted) standard conics are in scope. Finding the centre by intersecting the two asymptotes is intersection of lines, which is in Unit 10, so the explanation uses no out-of-unit tools.
- **ERR-219 U10-F-050** (E8): Errata ruling: translated (shifted) standard conics are in scope. Uses the focal-difference definition, midpoint and c^2 = a^2 + b^2 only.
- **ERR-304 U14-B-041** (E8): Editor ruling: solving trigonometric equations by general solution is out of scope and was removed (U14-B-014, U14-B-042). Counting intersections from graphs of trigonometric functions, and deciding whether a value lies in the range, test the graphs and ranges of trigonometric functions, which are in the syllabus. These items are retained.

## 5. Findings by unit

| Unit | Closures | With items changed |
|---|---|---|
| U01 | 9 | 2 |
| U02 | 29 | 13 |
| U03 | 22 | 7 |
| U04 | 22 | 16 |

| | | |
|---|---|---|
| U05 | 11 | 5 |
| U06 | 9 | 5 |
| U07 | 34 | 11 |
| U08 | 25 | 10 |
| U09 | 21 | 7 |
| U10 | 41 | 20 |
| U11 | 20 | 4 |
| U12 | 14 | 3 |
| U13 | 32 | 24 |
| U14 | 15 | 4 |

## 6. Automated test suite

25 of 25 automated tests passed on the finished release folder (2026-10-08 08:38 UTC).

| Test | Result |
|---|---|
| T01 schema: every record validates against question_bank_schema.json | Pass |
| T02 identity: 2,700 unique IDs, sequential per unit, unit counts as planned, local IDs unique | Pass |
| T03 options: 4 distinct options, key letter valid, key text equals keyed option, distinct after whitespace normalisation | Pass |
| T04 export parity: JSON = JSONL = CSV; student and teacher exports and question index agree with the bank | Pass |
| T05 KaTeX: every $...$ segment in every text field renders with throwOnError (strict parser) | Pass |
| T06 house style: no em/en dashes in any release text file; no control characters; one log convention; no letter references in solutions | Pass |
| T07 books: student books contain every unit ID and no keys, solutions or integer answers; teacher books contain every ID and every key | Pass |
| T08 rendered HTML: no raw LaTeX or stray $ in visible text of any book or sample page | Pass |
| T09 integer alternates: 342 present, each an exact non-negative integer, parent link correct, every non-equal value reviewed | Pass |

| | |
|---|---|
| T10 machine-check evidence: every recomputed record has a matching evidence entry with distinct distractor values and check source | Pass |
| T11 reproducibility: rebuilding every unit from the shipped authoring sources reproduces the released records exactly | Pass |
| T12 selection dependencies: family register matches records, every alias group links 2+ items, every errata alias request is enforced | Pass |
| T13 key balance: exact quotas per unit, no run longer than 3, no ABCD/DCBA run, no two units share a key sequence | Pass |
| T14 errata stability vs previous release: same IDs and key letters; unchanged items keep stem and option order; every change is in the closure register | Pass |
| T15 generator, full-bank mode: seeds 0-1004 each produce a paper that passes an independent validator; output is deterministic | Pass |
| T16 generator, unit-practice mode: seeds 0-199 for every unit pass the independent validator | Pass |
| T17 sample paper: sample files match seed 42 of the generator and the release summary | Pass |
| T18 checksums: every listed file matches; every non-KaTeX file is listed | Pass |
| T19 shipped verifier: verify_release.py (standard library only) exits 0 | Pass |
| T20 links: every link in START_HERE.html and the README file list resolves inside the release | Pass |
| T21 zip: archive opens, passes CRC, and every file in it matches the release folder byte for byte | Pass |
| T22 duplicates: no two stems identical; near-identical stems (character 5-gram Jaccard >= 0.85, same unit) are dependency-linked or have different answers | Pass |
| T23 option-length cue: the key is the strictly longest option in no more than chance (25%) of items | Pass |
| T24 scope sweep: no stem uses a topic excluded under the strict NTA 2026 reading | Pass |
| T25 summary consistency: release_summary, coverage_map and question_index agree with the records | Pass |

**Mutation check.** To show the suite catches real defects, three were injected into a copy of the release: record 6: key text replaced by malformed LaTeX ($\frac{1}{$); record 10: explanation ends with 'Hence option (B).'; Unit 3 student book: an 'Answer: (A)' paragraph inserted into the first article. Each injected defect was caught by the test aimed at it (T03, T05, T06, T07) and by the integrity tests (T04, T11, T18, T19).

## 7. Closure register

Every finding and how it was closed. The same data is in `Errata_Closure_Register.csv`.

| No. | Item | Cat. | Severity | Finding | Closure | Items changed |
|---|---|---|---|---|---|---|

| | | | | | | |
|---|---|---|---|---|---|---|
| ERR-001 | U01-B-006 JEE-MATH-U01-053 | E6 | major | Gives away the answer of U01-048 (and vice versa): 053 is the exact inverse of 048 on the same data, n(X)=3, n(Y)=4, 2^12 = 4096 relations from X to Y. The two items are in different concept families (U01-B-count-relations-reverse vs U01-B-count-relations-basic) with no shared family_dependencies entry, so the generator can put both in one paper. | **rewritten** Confirmed: U01-B-006 was the exact inverse of U01-B-001 (n(X)=3, n(Y)=4, 2^12). Rewritten with new data and a different unknown: n(X)=n(Y)+1, relations X to Y = 2^20 gives n(Y)=4, n(X)=5 via a quadratic; asks k where relations on X = 2^k times relations on Y (k=9). New family U01-B-relations-ratio-sizes; numerical alternate updated; distractors 1, 41, 5 each from a named slip. | U01-B-006 |
| ERR-002 | U01-C-040 JEE-MATH-U01-134 | E6 | major | Statement II of 134 does the same task as U01-133 (count f with f(f(x)) = x, i.e. involutions) with only the set size changed (3 instead of 4). 133's solution method (identity + single swaps + double swaps) settles 134 Statement II at once. Different families with no shared dependency. | **rewritten** Statement II (involution count on 3 elements) duplicated 133's task. Rewrote the item on a 4-element set: Statement I idempotent count f o f = f (41, true), Statement II number of f with f o f constant (true 40, claimed 92, false). Key unchanged (I true, II false); new family U01-C-idempotent-constant-square. Brute-force verified. | U01-C-040 |
| ERR-003 | U01-C-010 JEE-MATH-U01-104 | E6 | minor | Near-duplicate of U01-098: both are Easy 'find f^{-1} of the bijection f: R->R, f(x) = (power of x) + 2', with the same distractor set (sign slip x+2, wrong order of undoing, reciprocal 1/f). Not linked by family_dependencies. | **aliased** Related task (inverse of a different bijection), not the same data or answer; alias with U01-C-004 requested. | - |
| ERR-004 | U01-C-016 JEE-MATH-U01-110 | E6 | minor | Near-duplicate of U01-103: both ask 'the number of onto functions from {1,...,m} to an n-letter set' with only m, n changed (3->2 vs 4->3). Not linked by family_dependencies. | **aliased** Same method (onto count) with different sizes; brute force confirms 6 and 36. Alias with U01-C-009 requested. | - |
| ERR-005 | U01-C-015 JEE-MATH-U01-109 | E7 | minor | difficulty_reason gives the wrong number: it says 'Factorising $64$ as a power', but the stem's count of functions is 256. | **fixed** difficulty_reason now says 'Factorising $256$'. | - |
| ERR-006 | U01-C-021 JEE-MATH-U01-115 | E8 | minor | The explanation proves f(x) = x^3 - 3x is onto using limits at infinity and continuity (an intermediate-value argument). The U01 scope brief puts limits and continuity in Unit 7. The key (onto but not one-one) is correct. | **fixed** Onto argument now uses conjugate pairs of non-real roots of x^3 - 3x - c = 0 (real coefficients), no limits/continuity. Key unchanged (onto, not one-one; sympy range = R). | - |
| ERR-007 | U01-C-032 JEE-MATH- | E5 | minor | The distractor rationale for option A does not stand alone. It says 'Applies both the above errors', which refers to the rationales of other options and assumes a display order that the rationale list does not have (A is listed first). | **fixed** Rationale for 'neither one-one nor onto' made self-contained, naming both slips explicitly. | - |

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|---|---|---|---|---|---|---|
| | JEE-MATH-U01-126 | | | | | |
| ERR-008 | U01-B-015 JEE-MATH-U01-062 | E5 | minor | The distractor rationale for option B does not produce option B. 'Treats the listed pairs as unordered ... and misjudges transitivity' is vague: if the pairs are read as unordered, the relation becomes the symmetric closure, and in that relation the chain (2,1),(1,3) needs (2,3), which is present as {3,2}. So the transitive judgement does not follow from the stated error. | **fixed** Confirmed with enumeration: reading pairs as unordered makes R the full relation (all three properties), not option 'reflexive and symmetric but not transitive'. Rationale rewritten to a concrete slip that yields exactly that option: swapping the symmetry and transitivity definitions (chain test passes, reversal test fails on (2,1)). | - |
| ERR-009 | U01-C-003 JEE-MATH-U01-097 | E9 | minor | The subtopic metadata writes maths outside $...$ and uses set names A, B, while the item uses P, Q. | **fixed** Subtopic now 'Onto functions impossible when $n(P) < n(Q)$'. | - |
| ERR-010 | U02-B-036 JEE-MATH-U02-091 | E6 | major | Near-duplicate of JEE-MATH-U02-069 and not linked by family_dependencies (U02-omega-power-sum vs U02-omega-product). Both items use exactly the same two brackets, $1 - \omega + \omega^2$ and $1 + \omega - \omega^2$. The whole work of U02-091 is reducing them to $-2\omega$ and $-2\omega^2$, which is the full solution of U02-069. The generator could put both on one paper. | **rewritten** Confirmed: U02-B-036 reused the exact brackets of U02-B-014 ($-2\omega$, $-2\omega^2$). Rewritten as a counting question: number of $n$ in $1..20$ with $(1 + \omega)^n + (1 + \omega^2)^n = 2$ (key 3; distractors 6, 10, 7 from named slips, all brute-force verified). New family U02-omega-power-count; Medium kept; numerical alternate updated to 3. | U02-B-036 |
| ERR-011 | U02-B-022 JEE-MATH-U02-077 | E6 | minor | Same task as JEE-MATH-U02-071 (least distance from a fixed point to a circle, $\|z_0 - c\| - r$) with different numbers. They are in different families (U02-extremal-point-circle vs U02-extremal-circle), so U02-071/U02-076 and U02-077 can appear together. | **aliased** Same method (distance to centre minus radius) as U02-B-016/U02-B-021 with different numbers; related task, not a duplicate. Alias requested. | - |
| ERR-012 | U02-A-008 JEE-MATH-U02-008 | E6 | minor | Near-duplicate of JEE-MATH-U02-002 with no family link. U02-002 asks for the inverse of $3 + 4i$ (as a pair) and U02-008 for the inverse of $(2 - i)^2 = 3 - 4i$. The answers are conjugates with the same denominator $25$, so either item's working gives the other's answer. | **rewritten** Confirmed: inverse of (2 - i)^2 = 3 - 4i is the conjugate of U02-002's answer over the same denominator 25. Rewritten with base (3 - i)^2 = 8 - 6i and a different unknown, \|w\| for w = z^{-1} (key 1/10, via \|z^{-1}\| = 1/\|z\| = 1/\|3 - i\|^2); new distractors 1/sqrt(10), 1/100, 1/(2 sqrt 7), each from a named slip; new family U02-A-inverse-modulus. Still Easy, XI. | U02-A-008 |
| ERR-013 | U02-A-033 JEE-MATH-U02-033 | E6 | minor | The first step of U02-033 is to simplify $\frac{1+i}{1-i} = i$. That is the key computation, with the same quotient, of JEE-MATH-U02-020 (and of the linked U02-021 family). There is no family link, so one item gives away the core step of the other. | **aliased** Related task only: U02-033 asks for square roots, U02-020/021 ask for a least n; the shared step (1+i)/(1-i) = i is a common sub-step, not the same task or answer. Alias group requested with U02-A-020 and U02-A-021. | - |

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|---|---|---|---|---|---|---|
| ERR-014 | U02-B-054 JEE-MATH-U02-109 | E6 | minor | The key step of U02-109, $\|z - 1\|^2 + \|z + 1\|^2 = 4$ on $\|z\| = 1$, is the same identity with the same numbers as JEE-MATH-U02-079, whose answer states that $\|z-1\|^2 + \|z+1\|^2 = 4$ is the circle $\|z\| = 1$. There is no family link. | **aliased** U02-B-054 key step $\|z-1\|^2 + \|z+1\|^2 = 4$ on $\|z\| = 1$ is the U02-B-024 identity with the same numbers, but U02-B-054 asks a different extremal question; alias requested with U02-B-024. | - |
| ERR-015 | U02-B-051 JEE-MATH-U02-106 | E6 | minor | U02-106's key insight is that $\|z - a\| + \|z - b\| = \|a - b\|$ is the segment $ab$, not an ellipse. JEE-MATH-U02-090 tests and explains exactly that, with an unlinked family (U02-segment-locus-extremal vs U02-locus-degenerate). Seeing U02-090 on the same paper gives away the only non-routine step of U02-106. | **aliased** U02-B-035 gives away the degenerate-segment insight used in U02-B-051; related task, alias requested. | - |
| ERR-016 | U02-A-047 JEE-MATH-U02-047 | E6 | minor | U02-047's solution tabulates $S_n = i + \dots + i^n$ by $n \bmod 4$ ($i, -1+i, -1, 0$). This directly gives the answer to JEE-MATH-U02-004 ($S_{2027}$, $2027 \equiv 3$, so $-1$) and the first step of U02-052. U02-047 is not linked to U02-A-sum-powers-i. | **aliased** Confirmed overlap with U02-004 (residue table of S_n gives S_2027 = -1). Different unknown (a count), so aliased rather than rewritten: alias group U02-A-004, U02-A-047. U02-052 was rewritten for its E7 finding and no longer uses the S_n table. | U02-A-052 |
| ERR-017 | U02-A-034 JEE-MATH-U02-034 | E3 | minor | The explanation lists the squares of the three wrong options in an order that does not match the options. Read in option order (A, C, D), it gives the wrong square for each one. | **fixed** Verified with sympy: (y + ix)^2 = -a + ib, (y - ix)^2 = -a - ib, (x + iy)^2 = a + ib. Explanation now pairs each expression with its square by value, without option positions or letters. | - |
| ERR-018 | U02-A-033 JEE-MATH-U02-033 | E5 | minor | The rationale for distractor B does not produce $\pm(1 + i)$. 'Solves $x^2 - y^2 = 0$ as $x = y$ and takes the root of the numerator $1 + i$ itself' is not a coherent route: $(1+i)^2 = 2i$, and $1 + i$ is not a square root of $1 + i$. | **fixed** Confirmed: the old rationale did not produce +-(1 + i). Rationale now: simplifies the quotient as (1 + i)^2 = 2i, forgetting to divide by (1 - i)(1 + i) = 2; square roots of 2i are +-(1 + i) (sympy verified). Explanation already correct. | - |
| ERR-019 | U02-A-052 JEE-MATH-U02-052 | E7 | minor | Rated Difficult, but the item is three routine steps: reduce a sum of powers of $i$ modulo 4, do one complex division and take a squared modulus. No trap goes beyond the Medium items in the slice (e.g. U02-029 and U02-022, which are rated Medium). | **fixed** Kept Difficult to preserve batch quotas (17/25/13) and made the item genuinely harder: the sum is now sum_{k=1}^{30} i^{k^2}, where the exponent has only residues 0 and 1 mod 4 (trap: applying the period-4 cancellation of i^k). Sum 15 + 15i, x + iy = 3 + 9i, x^2 + y^2 = 90 (sympy). Distractors 2/5 (reads i^{k^2} as i^k), 450 (divides by \|2 - i\| not \|2 - i\|^2), 18 (divides by \|2 - i\|^4). Numerical alternate now asks x^2 + y^2 = 90. New family U02-A-sum-square-powers-equality. | U02-A-052 |
| ERR-020 | U02-D-003 JEE- | E6 | major | U02-148 Statement I gives away the answer to U02-168. 148 states and proves (p = -4, q = 1 in its explanation) that the rational-coefficient monic quadratic with root 2 + sqrt(3) has q = 1. 168 asks for exactly that equation, | **rewritten** Confirmed: U02-C-038 Statement I derives q = 1 for root 2 + sqrt(3), giving away U02-D-003's answer. Rewrote U02-D-003 as a different task: monic rational-coefficient | U02-D-003 |

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|---|---|---|---|---|---|---|
| | MATH-U02-168 | | | x^2 - 4x + 1 = 0. The items have different families (U02-statement-surd-conj vs U02-form-given-roots), so the generator can place both in one paper. | quadratic with root (sqrt(3) - 1)^2 = 4 - 2sqrt(3), so the root must first be expanded; key x^2 - 8x + 4 = 0 (sympy-verified), distractors from sign slip, dropped middle term (x^2 - 8x + 13), and product slip (x^2 - 8x + 28). New family U02-form-squared-surd-root; still Easy. | |
| ERR-021 | U02-D-051 JEE-MATH-U02-216 | E6 | minor | Four items reuse x^2 - 4x + 1 = 0 (168, 180, 204, 216). U02-180 Statement I states alpha^2 + beta^2 = 14 for this equation, and 14 = S_2 is exactly U02-216's answer. 180 also repeats the tasks of 166 (1/alpha + 1/beta) and 167 (alpha^2 + beta^2) with trivially different numbers. None of these share a family_dependencies entry. | **rewritten**<br>Confirmed: U02-D-015 Statement I states alpha^2 + beta^2 = 14 for x^2 - 4x + 1 = 0, which is exactly U02-D-051's answer S_2. Since one item gives the other's answer directly, rewrote U02-D-051: x^2 - 5x + 1 = 0, find (S_12 - S_8)/(S_11 - S_9) = 5 via the recurrence at n = 11 and n = 9 (sympy-verified). New family U02-recurrence-difference-ratio; still Difficult; numerical alternate updated to 5. | U02-D-051 |
| ERR-022 | U02-C-015 JEE-MATH-U02-125 | E6 | minor | U02-125 (range of x^2 - 4x + 7, answer [3, inf)) and U02-116 (minimum of x^2 - 6x + 13, answer 4) are the same task with trivially different numbers: complete the square on a monic quadratic and report the vertex value. They are in different families (U02-range-quadratic vs U02-min-quadratic), and both are Easy single-step items. | **aliased**<br>Verified: both are complete-the-square vertex value on a monic quadratic with different data and different asks (minimum vs range); related task, not a repeat, so aliased U02-C-015 with U02-C-006. | - |
| ERR-023 | U02-D-013 JEE-MATH-U02-178 | E6 | minor | U02-178 (form the real-coefficient monic quadratic with root 3 - 2i, giving x^2 - 6x + 13) and U02-115 (2 - 3i is a root of x^2 + px + q with p, q real; find q = 13) are the same conjugate-pair-plus-product task with swapped digits, and both give 13 as the product. They are in different families (U02-form-given-roots vs U02-conjugate-pair). | **aliased**<br>Confirmed related task (conjugate complex root, product 13) but different data and unknown (full equation vs q). Alias U02-D-013 with U02-C-005. | - |
| ERR-024 | U02-D-040 JEE-MATH-U02-205 | E6 | minor | U02-205 is in effect the sum-of-cubes task of U02-171/183 (family U02-sum-cubes) with trivially different numbers. Since alpha*beta = -1, the denominator (alpha*beta)^2 = 1, so the item reduces to alpha^3 + beta^3 for x^2 - 3x - 1 = 0, next to 171's x^2 - 3x - 2 = 0. Its family is separate (U02-cubes-over-square-product). | **aliased**<br>Confirmed: with alpha*beta = -1 the item reduces to alpha^3 + beta^3 = 36, the sum-of-cubes task. Different data and answer from U02-D-006 (45). Alias U02-D-040 with the U02-sum-cubes family items U02-D-006 and U02-D-018. | - |
| ERR-025 | U02-C-044 JEE-MATH-U02-154 | E6 | minor | U02-154 repeats U02-139's task (the range of (ax^2 - bx + c)/(ax^2 + bx + c) by the discriminant method, with y = 1 attained at x = 0) with trivially different numbers, plus a final counting step. The families differ (U02-range-rational-count vs U02-range-rational). | **aliased**<br>Verified: same discriminant-method structure as U02-C-029 with different coefficients plus an integer count; related, not the same data or answer, so aliased U02-C-044 with U02-C-029. | - |
| ERR-026 | U02-D-035 JEE-MATH-U02-200 | E6 | minor | U02-200 Statement I is the task of U02-169 (equation with reciprocal roots, obtained by reversing the coefficients) with different numbers, in a different family (U02-transform-statement vs U02-transform-reciprocal). | **aliased**<br>Confirmed: Statement I repeats the reciprocal-roots transformation of U02-D-004 with different numbers; related task only. Alias U02-D-035 with U02-D-004. | - |

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| ERR-027 | U02-D-037 JEE-MATH-U02-202 | E6 | minor | U02-202 Statement II (S_4 for x^2 - 2x - 1 = 0 by iterating S_n = 2S_{n-1} + S_{n-2}) is the same task as U02-201 (S_5 for x^2 - x - 1 = 0 by iterating S_n = S_{n-1} + S_{n-2}) with trivially different numbers. The families differ (U02-recurrence-statement vs U02-recurrence-value). | **aliased** Confirmed: Statement II iterates the power-sum recurrence like U02-D-036 with different equation and index; related task only. Alias U02-D-037 with U02-D-036. | - |
| ERR-028 | U02-C-024 JEE-MATH-U02-134 | E5 | minor | The rationale for distractor C does not produce its option. Expanding (k - 3)^2 with the sign slip k^2 + 6k + 9 and then subtracting 4k gives k^2 + 2k + 9, which is never negative, so it yields no interval at all. It does not give k^2 + 10k + 9 or (-9, -1). | **fixed** Verified with sympy: the named slip gives k^2 + 2k + 9 (D = -32), never negative. Rationale rewritten: correct k^2 - 10k + 9 < 0 factorised as (k + 1)(k + 9), giving exactly (-9, -1). | - |
| ERR-029 | U02-C-038 JEE-MATH-U02-148 | E5 | minor | The rationale for distractor B is muddled. It says the student 'computes the product as 4 - 3 incorrectly', but 4 - 3 = 1 is the correct product (2 + sqrt3)(2 - sqrt3). This does not describe a path to rejecting Statement I. | **fixed** Verified: 4 - 3 = 1 is correct. Rationale rewritten: product computed as 4 + 3 = 7, so Statement I rejected, Statement II correctly rejected, giving 'Both false'. | - |
| ERR-030 | U02-C-023 JEE-MATH-U02-133 | E5 | minor | The stem states the trap condition outright ('is a quadratic equation (so k != 0)'). Distractor A, (-inf, 4], contains k = 0 and so is ruled out on sight. The item's stated misconception ('Forgetting the a != 0 condition') can no longer catch anyone. | **fixed** Verified: parenthetical gave away k != 0. Stem reworded to 'If kx^2 + 4x + 1 = 0 is a quadratic equation with real roots, then the set of all possible values of k is'; option (-inf, 4] now a live trap. | U02-C-023 |
| ERR-031 | U02-C-032 JEE-MATH-U02-142 | E5 | minor | Distractor D, [1, 4], contains k = 1, which the stem explicitly excludes ('Let k != 1'), so it is ruled out on sight. Only three options really compete. | **fixed** Verified: [1, 4] contains the excluded k = 1. Replaced with (1, 4] (product <= 0 slip, accepts k = 4 where the roots are 0 and -1); value updated. | U02-C-032 |
| ERR-032 | U02-C-010 JEE-MATH-U02-120 | E5 | minor | Distractors A, B and D all use one slip (b^2 + 4ac). Under that slip every option, including the key C (25 + 12 = 37), gives a non-square, so the modelled error does not lead to any single wrong choice. The rationales therefore do not explain why a student would pick A, B or D. | **fixed** Verified: under b^2 + 4ac every option was non-square. Replaced all four equations: key 2x^2 - 5x - 6 (D = 73); distractors 2x^2 - 7x + 6 (only non-square under b^2 + 4ac), x^2 - 4x - 12 (only under b^2 - 2ac), 3x^2 - x - 4 (only under b^2 - 4c). Under each slip exactly one option looks irrational and the key looks rational. Check, explanation, misconception updated. | U02-C-010 |
| ERR-033 | U02-C-017 JEE-MATH- | E7 | minor | Topic taxonomy is inconsistent. U02-127, 128 and 129 (and 131) are tagged topic 'Quadratic equations' with subtopic 'Quadratics with complex coefficients' or similar. U02-173, 181, 191 and 217 use a separate top-level | **fixed** Moved U02-C-017, 018, 019 and 021 to topic 'Quadratics with complex coefficients' (matching U02-D); subtopics of 017/018 now 'Solving with a complex discriminant'. | - |

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| | U02-127 | | | topic 'Quadratics with complex coefficients'. Topic-based filtering in the generator will split one concept across two topics. | | |
| ERR-034 | U02-C-031 JEE-MATH-U02-141 | E9 | minor | The stems of U02-141, 152 and 160 open by stating the condition as a fact ('Both roots of ... are real and positive (the roots may be equal).') and then ask for 'The set of all such real k'. A student can misread this as saying the roots are real and positive for every k. Standard JEE wording puts the condition inside the question. | **fixed** Stem reworded to 'The set of all real k for which both roots of ... are real and positive (possibly equal) is'; same change applied to U02-C-042 (U02-152) and U02-C-050 (U02-160). | U02-C-031; U02-C-042; U02-C-050 |
| ERR-035 | U03-A-008 JEE-MATH-U03-008 | E6 | major | Near-duplicate of JEE-MATH-U03-141 (other half of unit 3): same task with trivially different numbers, not linked by family_dependencies (U03-A-involutory-param vs U03-C-involutory-param). | **rewritten** Confirmed near-duplicate of U03-C-031 (both reduce P^2 = I to (9 - ck)I = I). Rewrote U03-A-008 (still Easy, Inverse/parameter): P = [[a, 4], [-6, b]], P^2 = I, find ab; the off-diagonal entries force b = -a before the diagonal gives a^2 = 25, so ab = -25 (sympy solve: (5, -5), (-5, 5)). New distractors 25, -23, -24 each from a named slip; new family U03-A-involutory-trace-zero. | U03-A-008 |
| ERR-036 | U03-A-019 JEE-MATH-U03-019 | E6 | major | Uses the same matrix P = [[1, 2], [0, 1]] as JEE-MATH-U03-036 and the same key fact P^n = [[1, 2n], [0, 1]]; the two items give each other away if they appear in one paper, yet they are in different families with no shared dependency. | **rewritten** Confirmed: U03-A-019 and U03-A-036 used the same matrix [[1, 2], [0, 1]] and the same fact P^n = [[1, 2n], [0, 1]]. Rewrote U03-A-019 (still Medium): P = [[2, 1], [0, 2]] = 2I + N, P^n = [[alpha, beta], [0, alpha]], beta = 6 alpha gives n/2 = 6, n = 12 (verified by sympy loop). Different matrix, binomial scalar-plus-nilpotent method and a ratio twist; new family U03-A-power-scalar-plus-nilpotent; numerical alternate updated to 12. | U03-A-019 |
| ERR-037 | U03-B-015 JEE-MATH-U03-070 | E6 | minor | Near-duplicate of JEE-MATH-U03-089: both statement items test exactly \|P^T\| = \|P\| and the sign of \|kP\| for order 3 (k = -1 vs k = -2); not linked by family_dependencies. | **aliased** Related but not the same task: U03-B-015 tests the general facts (\|P^T\| = \|P\|, sign of \|-P\|); U03-B-034 uses numerical data (\|-2P\| with \|P\| = 3, \|P^T P\|). Different data and answers, so no rewrite; alias link requested. | - |
| ERR-038 | U03-A-039 JEE-MATH-U03-039 | E6 | minor | U03-039 teaches and asks for the identity tr(P^2) = -2(p12^2 + p13^2 + p23^2) for 3x3 skew-symmetric P, which is the entire key step of JEE-MATH-U03-051; families not linked. | **aliased** Confirmed related, not duplicate: U03-A-039 computes tr(P^2) = -2(sum of squares) for given entries, U03-A-051 counts matrices using that identity (sympy: tr(P^2) = -2x^2 - 2y^2 - 2z^2). Alias requested so they never share a paper. | - |
| ERR-039 | U03-A-013 JEE-MATH-U03-013 | E5 | minor | Distractor rationale for option C is self-contradictory: it says the student 'swaps the two facts' but then states both matrices are believed skew-symmetric. | **fixed** Rationale for the 'Statement I false, Statement II true' distractor reworded: student thinks P + P^T is skew-symmetric (Statement I false) and P P^T is skew-symmetric (Statement II true), confusing both with P - P^T. Stem and options unchanged. | - |

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| ERR-040 | U03-B-015 JEE-MATH-U03-070 | E5 | minor | Distractor rationale for option D does not describe a concrete error that produces the option. | **fixed** Rationale for 'Statement I is false but Statement II is true' now names the slips (believes transpose changes sign; uses (-1)^2). Rationale for 'Both false' also made explicit. | - |
| ERR-041 | U03-A-041 JEE-MATH-U03-041 | E3 | minor | Explanation asserts that P^2 = P with P not O or I forces a + d = 1 and ad - bc = 0, but the entry equations it quotes do not give this in the case b = c = 0, and ad - bc = 0 is never derived. | **fixed** Count 12 re-verified by brute force over 81 matrices. Explanation now derives the case split: if a + d != 1 then b = c = 0, a = d in {0, 1}, giving only O and I; if a + d = 1 then bc = a - a^2 = ad = 0 for entries in {0, 1, 2}, giving 5 + 5 = 10 matrices; total 12. | - |
| ERR-042 | U03-B-040 JEE-MATH-U03-095 | E3 | minor | Column operations are listed in an order that, applied as written, does not produce the displayed matrix. | **fixed** Verified: stated order gives (4, 5, 11). Explanation now says C3 -> C3 - C2 and then C2 -> C2 - C1 (and orders the row operations explicitly). Value -8 unchanged. | - |
| ERR-043 | U03-B-046 JEE-MATH-U03-101 | E3 | minor | Row operations are listed in an order that, applied as written, does not give the stated rows. | **fixed** Verified. Explanation now says R3 -> R3 - R2 and then R2 -> R2 - R1. Value unchanged; misconception LaTeX tidied. | - |
| ERR-044 | U03-B-021 JEE-MATH-U03-076 | E8 | minor | The item is in scope, but its explanation relies on linear-algebra ideas ('two-dimensional span', 'dependent' vectors) that are outside the NCERT/NTA syllabus, which excludes rank. | **fixed** Span/dependence argument replaced by the zero-padding factorisation PQ = [P\|0] [Q;0] into 3x3 factors with zero determinant, plus the explicit PQ; checked in sympy. Difficulty reason updated. | - |
| ERR-045 | U03-B-039 JEE-MATH-U03-094 | E9 | minor | Maths outside $...$ in student-facing metadata text. | **fixed** Wrapped $x$ and $R_1 \to R_1 - 2R_2 + R_3$ in LaTeX. | - |
| ERR-046 | U03-B-029 JEE-MATH- | E9 | minor | Maths outside $...$ in common_misconception. | **fixed** Wrapped $x = 0$. | - |

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| | U03-084 | | | | | |
| ERR-047 | U03-B-048 JEE-MATH-U03-103 | E9 | minor | Maths outside $...$ in common_misconception. | **fixed** Wrapped $(a+b+c)$. | - |
| ERR-048 | U03-C-031 JEE-MATH-U03-141 | E6 | major | Near-duplicate of JEE-MATH-U03-008 (first half of unit 3). Both give a 2 by 2 matrix with diagonal 3, -3 and one unknown off-diagonal entry and impose P^2 = I (141 phrases it as P^{-1} = P, which is the same condition). Each reduces to 9 + (product of off-diagonal entries) = 1, i.e. off-diagonal product = -8. Families U03-A-involutory-param and U03-C-involutory-param share no family_dependencies entry, so a generated paper can contain both. | **rewritten** Confirmed near-duplicate of U03-A-008 (involutory 2x2, off-diagonal product -8). U03-C-031 rewritten: abstract real 2x2 P with P^{-1} = P, P != +-I; using the stated identity P^2 - (tr P)P + \|P\|I = O deduce tr P = 0, \|P\| = -1, then \|adj P + 3P^{-1}\| = \|2P\| = -4. New family U03-C-involutory-adj-det; sympy check solves P^2 = I generically and confirms the value is -4 for every non-scalar solution. Difficulty Medium kept; qtype now Conceptual/application MCQ. Note U03-A-008 was itself independently rewritten by batch A (trace-zero family), no overlap with the new item. | U03-C-031 |
| ERR-049 | U03-D-048 JEE-MATH-U03-213 | E6 | major | JEE-MATH-U03-187 gives away the key step of 213. Both use the same coefficient matrix (rows (1,1,1),(1,2,3),(1,3,5)). 187's answer is 'consistent iff a, b, c are in A.P.', so in a paper containing both, 213 collapses to solving 1 + cos 2θ = 2 sin θ at sight. Families U03-D-consistency-constants and U03-D-consistency-trig are not linked. | **rewritten** Confirmed: 048 used the same rows (1,1,1),(1,2,3),(1,3,5) as U03-D-022, so 022's A.P. answer gave away its key step. 048 also solved a trigonometric equation on an interval, which is out of scope. Rewritten as a new item: x + 2y - z = p, 2x - y + 3z = q, 4x + 3y + z = pq - 4 with p, q positive integers. R3 = 2R1 + R2 gives (p - 1)(q - 2) = 6, so there are 4 ordered pairs. New family U03-D-consistency-integer-pairs. Difficult kept. Sympy check finds the left null vector and scans the pairs; numerical alternate is 4. | U03-D-048 |
| ERR-050 | U03-D-054 JEE-MATH-U03-219 | E6 | major | JEE-MATH-U03-200 gives away 219. Statement I of 200 states that X1 + t(X2 - X1) solves PX = Q for every real t when X1, X2 are solutions; 219 asks which vector is necessarily a solution given two solutions, and the key is exactly X1 + t(X2 - X1) with t = -1. Families U03-D-concept-solution-set and U03-D-linearity are not linked. | **rewritten** Confirmed: Statement I of U03-D-035 states X1 + t(X2 - X1), and 054's key was that formula with t = -1. Rewritten as a new item: P = [[1,1,1],[1,2,alpha],[2,beta,4]] with solutions (1,1,1) and (2,-1,2) of PX = Q. P(X2 - X1) = O gives alpha = beta = 3. The row relation R3 = R1 + R2 then makes PX = (2,k,7)^T consistent iff k = 5. Distractors: 9/2 (column relation applied to the RHS), 9 and -5. New family U03-D-solutions-fix-matrix. Difficult kept; qtype is now Inverse/parameter MCQ. Sympy check solves for alpha and beta and runs a rank test. | U03-D-054 |
| ERR-051 | U03-D-021 JEE-MATH-U03-186 | E6 | major | JEE-MATH-U03-149 gives away the determinant work of 186 (and of 216). 149 asks for the k where [[k,1,1],[1,k,1],[1,1,k]] is singular (key {1, -2}); 186's coefficient matrix is exactly this matrix and 216's augmented determinant is the same determinant. With 149 in the same paper, 186 and 216 reduce to checking the two given roots. 149 (U03-C-singular-param), 186 (U03-D-cyclic-k) and 216 (U03-D-2var-3eqn) share no family_dependencies entry. | **aliased** Confirmed overlap: the determinant (k-1)^2(k+2) is shared by U03-C-039 (singular values {1,-2}), U03-D-021 and U03-D-051. This is a related task, not a copy. 021 still needs the constants tested at each root (only k = -2 gives no solution), and 051 is a two-variable concurrency sum. Alias requested for the group. qlib has no family_dependencies field, so the link is made through alias_requests. | - |
| ERR-052 | U03-C-045 | E6 | minor | 155 is the same task as JEE-MATH-U03-152 (recover a 3 by 3 matrix from its adjoint via adj(adj Q) = \|Q\| Q, i.e. Q = \|Q\| (adj Q)^{-1}) with an extra | **aliased** Confirmed: 045 recovers Q via adj(adj Q) = \|Q\|Q, same core task as 042 (P = \|P\|(adj | - |

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| | JEE-MATH-U03-155 | | | parameter step, but 155 is filed under U03-C-adj-det-param (with 139) and is not linked to 152's family U03-C-A-from-adj. Two 'matrix from its adjoint' items can therefore appear in one paper. | P)^{-1}) with an extra parameter step. Related, not a copy (different data, unknown is a trace). Alias requested [U03-C-045, U03-C-042]; qlib has no family_dependencies field, so linkage is via alias_requests. | |
| ERR-053 | U03-D-015 JEE-MATH-U03-180 | E6 | minor | 180 and JEE-MATH-U03-167 are the same task with different numbers: a homogeneous 3 by 3 system with k in one entry of the third row, where Δ is linear in k. 167 asks for the k giving non-trivial solutions; 180 asks for the condition for only the trivial solution, which is the complement of the same computation. The families U03-D-homog-nontrivial and U03-D-homog-trivial are not linked. | **aliased**<br>Confirmed: 015 (only the trivial solution, k != 10) is the complementary criterion of 002 (non-trivial solution, k = 2), with Delta linear in k in both. It is a related task with different data and a different answer. Alias requested [U03-D-015, U03-D-002]. | - |
| ERR-054 | U03-C-044 JEE-MATH-U03-154 | E2 | critical | The stem adds 'non-symmetric' to rule out coincidences among the options, but that does not rule them out. For a non-symmetric orthogonal P (which is non-singular and satisfies PP^T = P^TP), options A (Q^2), B (I) and D (PP^T) all equal QQ^T. B is the only option that holds for every admissible P, but 'Then QQ^T equals' does not say 'for every such P', so a student can defend A or D. | **fixed**<br>Confirmed with sympy: for P = [[0,1,0],[-1,0,0],[0,0,1]] (non-symmetric, orthogonal, normal) Q^2 = PP^T = QQ^T = I. Stem now says 'real' and 'for every such P, QQ^T necessarily equals'; distractors PP^T and Q^2 replaced by -I and Q^T. Options now I, -I, Q, Q^T: Q = I iff P symmetric (excluded), so Q != I and Q^T = Q^{-1} != I; QQ^T has non-negative diagonal so never -I. Exactly one option holds for every admissible P. Rationales, values and explanation updated. | U03-C-044 |
| ERR-055 | U03-D-054 JEE-MATH-U03-219 | E3 | minor | The explanation lists what the other options give as '2Q, O and 2Q', which does not match the option order A, B, C (A gives O, B gives 2Q, C gives 2Q). | **rewritten**<br>Confirmed: the explanation listed the distractor results out of option order. This is moot because 054 was fully rewritten (see the E6 closure). The new explanation does not refer to options by position. | U03-D-054 |
| ERR-056 | U03-D-007 JEE-MATH-U03-172 | E7 | minor | question_type is 'Single-step computation MCQ', but the item needs two 3 by 3 determinants (Δ and Δ3) and a division. The record's own difficulty_reason says 'Two $3 \times 3$ determinants and one division'. | **fixed**<br>Confirmed: the item needs Delta and Delta_3. question_type changed to Multi-step computation MCQ. Difficulty Easy kept, so the quotas are unchanged. | - |
| ERR-057 | U04-C-013 JEE-MATH-U04-120 | E6 | major | Same data set as U04-062 (5 men/boys, 4 women/girls, choose 3, at least one woman/girl); U04-120 is U04-062 times 3!, and its option B (74) is exactly U04-062's key. Families U04-B-at-least-complement and U04-C-complement-select-arrange are not linked. | **rewritten**<br>Confirmed: same data as U04-B-008 and 9P3-5P3 = 6 x 74. Rewritten to 6 men and 4 women seated in 3 chairs, complement of 'both end chairs men' (720 - 240 = 480); new family U04-C-complement-positions. | U04-C-013 |
| ERR-058 | U04-B-019 JEE-MATH- | E6 | major | Same data as U04-058 (committee of 5 from 6 men and 4 women); U04-058's key (exactly 2 women = 120) is one case of U04-073's direct count and appears as U04-073 option D. Families U04-B-committee-exact and U04-B-at-least-k are not linked. | **rewritten**<br>U04-B-019 rewritten as an inverse at-least-k problem: group of 3 from n men and 5 women, 80 groups with at least 2 women, find n (= 7). New data, new unknown, new family U04-B-at-least-k-inverse; distractors from dropping the all-women case (8), fill-minimum overcount (5), at-least-1 misreading (4). Numerical alternate updated. | U04-B-019 |

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| | U04-073 | | | | | |
| ERR-059 | U04-C-021 JEE-MATH-U04-128 | E6 | major | Same equation as U04-110 (non-negative solutions of x+y+z=10). U04-128's first step and its option B (66) are U04-110's key. Families U04-C-stars-bars-basic and U04-C-stars-bars-upper are not linked. | **rewritten** Confirmed: same equation as U04-C-003 (66). Rewritten to three coloured dice summing to 12 (lower and upper bounds together): 55 - 30 = 25; new family U04-C-stars-bars-two-sided; numerical alternate updated. | U04-C-021 |
| ERR-060 | U04-C-009 JEE-MATH-U04-116 | E6 | major | U04-115 and U04-116 use the same data (6 distinct books, 3 groups of 2), unlabelled vs labelled. Each item's key is an option in the other (115 option C = 90 = 116 key; 116 option C = 15 = 115 key), so one answers the other. Families U04-C-groups-unlabelled and U04-C-groups-labelled are not linked. | **rewritten** Confirmed: same 6 books / 3 pairs data as U04-C-008, keys mutually present. Rewritten to 8 books, 4 students, 2 each, atlas and dictionary with the same student: 4 x 90 = 360; new family U04-C-groups-labelled-pair. | U04-C-009 |
| ERR-061 | U04-C-045 JEE-MATH-U04-152 | E6 | major | U04-152 (5 distinct balls, 3 identical boxes, none empty) uses the same data as U04-129 (5 distinct balls, 3 distinct boxes, none empty); U04-152 option D = 150 is U04-129's key and the explanation states 25 = 150/3!. Families U04-C-distinct-onto and U04-C-groups-unlabelled-cases are not linked. | **rewritten** Confirmed: brute force 25 vs 150 on same data as U04-C-022. Rewritten to 7 distinct balls in 3 identical non-empty boxes with at least one box of exactly 3: 70 + 105 = 175; new family U04-C-identical-boxes-condition. | U04-C-045 |
| ERR-062 | U04-A-047 JEE-MATH-U04-047 | E6 | major | U04-047 (five-digit strings over {1,2,3} using every digit) is the same count as U04-129 (onto maps from 5 distinct balls to 3 distinct boxes): same task in a different costume with identical numbers and identical answer 150, solved by the identical inclusion-exclusion 3^5 - 3*2^5 + 3. Families U04-A-surjective-digits and U04-C-distinct-onto are not linked. | **rewritten** Verified: old item (5-digit strings over {1,2,3}, all used) = 150 = onto maps 5 balls to 3 boxes in U04-C-022. Rewritten to five-digit numbers using each of 0,1,2,3 at least once (key 180): four-symbol alphabet with leading-zero correction split into two cases; new family U04-A-every-digit-zero; distractors 240, 144, 360 recomputed; numerical alternate 180. | U04-A-047 |
| ERR-063 | U04-C-006 JEE-MATH-U04-113 | E6 | major | U04-113 (5 boys, 3 girls, no two girls adjacent) is U04-010 (4 boys, 3 girls, no two girls adjacent) with trivially different numbers, in unlinked families U04-A-gap-method and U04-C-gap-method. It also reuses U04-006's exact data (5 boys, 3 girls): U04-006 option C (14400) is U04-113's key, and U04-113 option C (36000) is U04-006's 'not all together' distractor. | **rewritten** Confirmed: repeats U04-A-006 data and U04-A-010 task. Replaced by 6 different novels and 3 identical dictionaries, no two dictionaries adjacent: 6! x 7C3 = 25200; new family U04-C-gap-mixed-identical. | U04-C-006 |
| ERR-064 | U04-A-036 JEE-MATH-U04-036 | E6 | major | U04-036 gives away U04-009 Statement II. U04-009 Statement II claims P(n,r) = P(n-1,r) + P(n-1,r-1); U04-036 shows the same identity with coefficient k and has key k = r. A student who sees both knows Statement II is false. Families U04-A-Pnr-statement and U04-A-Pnr-identity are not linked. | **rewritten** Verified: key k = r states the correct recurrence and so refutes U04-A-009 Statement II. Rewritten to a different unknown: P(m+n,2)=90, P(m-n,2)=30, find P(m,n) (key 56); new family U04-A-Pnr-system; sympy check solves the system; numerical alternate 56. | U04-A-036 |

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| ERR-065 | U04-C-033 JEE-MATH-U04-140 | E6 | major | U04-140 Statement II (3 numbers from {1,...,10}, no two consecutive) is U04-096 (3 numbers from {1,...,20}, no two consecutive) with trivially different numbers. Families U04-C-statement-stars-bars and U04-B-no-consecutive are not linked. | **rewritten** Confirmed: Statement II was the U04-B-042 no-consecutive task. Statement II replaced by a lower-bound stars and bars count (x+y+z=15, x>=3, y>=4, z>=5 gives 5C2 = 10, true); key unchanged, rationales and explanation updated. | U04-C-033 |
| ERR-066 | U04-B-012 JEE-MATH-U04-066 | E6 | major | U04-066 is made of two near-duplicates. Statement I (rectangles on a 4x4 board) is U04-065 (rectangles on an 8x8 board) with different numbers. Statement II (10 points, exactly 4 collinear, count lines) is U04-061 (12 points, 5 collinear, count lines) with different numbers, and it uses the same point set as U04-068. Its family U04-B-statement-grid-lines is linked to neither U04-B-grid-rectangles nor U04-B-lines-collinear. | **rewritten** U04-B-012 both statements replaced: I = triangles formed by 8 lines in general position (claimed 28, true 56); II = triangles through a fixed point among 9 concyclic points (claimed 36, true 28). No grid rectangles or collinear-points line count remains. Key stays both false. New family U04-B-statement-lines-circle; brute-force check. | U04-B-012 |
| ERR-067 | U04-B-038 JEE-MATH-U04-092 | E6 | major | U04-092 is made of two near-duplicates. Statement I (diagonals of a 20-gon) is U04-059 (diagonals of a 15-gon) with trivially different numbers. Statement II (divisors of 2^3*3^4 that are multiples of 6) is U04-078 (divisors of 10800 that are multiples of 6). Family U04-B-statement-diag-divisor is linked to neither U04-B-diagonals nor U04-B-divisor-multiples. | **rewritten** U04-B-038 both statements replaced, using N = 2^6 * 3^5: I = perfect-cube divisors (claimed 2, true 6); II = divisors that are squares or cubes (16, with the sixth-power overlap removed). Diagonal and multiples-of-6 tasks are gone. Key stays I false, II true. New family U04-B-statement-power-divisors; subtopic and cognitive level updated. | U04-B-038 |
| ERR-068 | U04-C-038 JEE-MATH-U04-145 | E6 | major | U04-145 (positive solutions of a+b+c+d = 10, in a coin costume) is U04-111 (positive solutions of x1+x2+x3+x4 = 12) with trivially different numbers. Families U04-C-selection-with-repetition and U04-C-stars-bars-basic are not linked. | **rewritten** Confirmed: positive solutions of a 4-variable sum, same as U04-C-004. Rewritten to 8 coins with unlimited Rs 1, 2, 5 and only two Rs 10 coins: 45 + 36 + 28 = 109; new family U04-C-selection-limited-supply. | U04-C-038 |
| ERR-069 | U04-C-011 JEE-MATH-U04-118 | E6 | minor | Positive-solution stars and bars is repeated, unlinked, across U04-111 (stars-bars-basic), U04-118 (identical-distribution, 8 chocolates among 3 children, each at least 1), U04-139 Statement I (9 identical balls into 4 boxes, none empty) and U04-144 Statement II (6 identical books, 2 children, each at least 1). Each is the same task C(n-1, k-1) with different numbers. | **aliased** Related task (positive stars and bars with different numbers and contexts), not a copy; alias group requested. | - |
| ERR-070 | U04-C-050 JEE-MATH-U04-157 | E6 | minor | U04-157 Statement II (9 distinct objects in three groups of 3, given to 3 persons) is the U04-116/U04-135 task (equal labelled groups) with different numbers, and U04-139 Statement II (4 balls in two unlabelled groups of 2) is the U04-115/U04-136 task. Family U04-C-statement-distribution is not linked to U04-C-groups-labelled or U04-C-groups-unlabelled. | **aliased** Statement items reuse the labelled / unlabelled equal-group tasks with different numbers; alias groups requested (U04-C-009 included since it is now a labelled equal-group item). | U04-C-009 |

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| ERR-071 | U04-B-015 JEE-MATH-U04-069 | E6 | minor | U04-069 Statement II (10C3 + 10C4 against 11C3) repeats U04-056's first step (10C4 + 10C5 = 11C5), and Statement I (20C7 = 20C13) repeats the symmetry used in U04-055. Family U04-B-statement-properties is not linked to U04-B-pascal-chain or U04-B-symmetry-equation. | **aliased**<br>Confirmed: Statement I uses the symmetry from U04-B-001 and Statement II the Pascal step from U04-B-002, with different numbers. Related tasks, not the same data, so the item is kept and an alias group is requested. | - |
| ERR-072 | U04-A-028 JEE-MATH-U04-028 | E6 | minor | U04-028 (BANANA, N's not adjacent) reuses U04-003's word, and its check line computes U04-003's key (60 arrangements in all). Families U04-A-gap-identical and U04-A-identical-word are not linked. | **fixed**<br>Verified: BANANA reused and the check line stated U04-A-003's key 60. Word changed to BALLOON (L's apart, key 900; check line 1260 - 360) so no U04-A-003 data or key appears; distractors 360, 1800, 540 recomputed; numerical alternate 900. | U04-A-028 |
| ERR-073 | U04-A-040 JEE-MATH-U04-040 | E6 | minor | U04-040's complement solution needs U04-002's key (648 three-digit numbers with distinct digits) as an intermediate step, and U04-040's explanation states it outright. Families U04-A-exactly-two-equal and U04-A-digits-distinct are not linked. | **aliased**<br>Verified: complement route uses 648 (U04-A-002 key). Related task, not the same task; alias requested for U04-A-040 with U04-A-002. | - |
| ERR-074 | U04-C-007 JEE-MATH-U04-114 | E6 | minor | U04-114 (3-letter codes over 6 letters with at least one repeat, by complement) is the same complement-of-all-distinct task as U04-026 (four-digit numbers with at least one repeated digit). Families U04-C-complement-repetition and U04-A-complement-repeat are not linked. | **aliased**<br>Same complement-of-all-distinct idea in a different setting; alias requested with U04-A-026. | - |
| ERR-075 | U04-A-049 JEE-MATH-U04-049 | E6 | minor | U04-049 Statement I (rank of SMALL) is the same task as U04-020/U04-044 (rank of a word with a repeated letter, family U04-A-rank-repeated) with a different word. U04-049's family U04-A-rank-ie-statement is not linked to it. | **aliased**<br>Verified: Statement I (rank of SMALL = 58) is the rank-with-repeated-letter task of U04-A-020/U04-A-044. Alias requested for U04-A-049 with U04-A-020 and U04-A-044. | - |
| ERR-076 | U04-C-001 JEE-MATH-U04-108 | E9 | minor | The unit uses three notations for the same quantities: batch A (U04-001 to 054) uses P(n, r) and C(n, r), batch B (U04-055 to 107) uses ${}^{n}C_{r}$, and batch C (U04-108 to 160) mixes \binom{n}{r} with ${}^{n}P_{r}$, sometimes in the same explanation (e.g. U04-120, U04-131, U04-156). JEE papers and NCERT use ${}^{n}C_{r}$ and ${}^{n}P_{r}$. | **fixed**<br>All batch C stems, explanations and rationales normalised to ${}^{n}C_{r}$ and ${}^{n}P_{r}$ (no \binom left). Batch A (P(n, r), C(n, r)) is outside this file and needs its own normalisation. | U04-C-013 |

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| ERR-077 | U04-A-004 JEE-MATH-U04-004 | E9 | minor | In U04-004, 012, 017, 035 and 036, the definition of P(n, r) is a parenthetical tacked on after the stem's final 'equals'/'is', so the stem reads '... then $n$ equals ($P(n, r)$ denotes ...)'. On screen this looks like part of the answer blank. | **fixed** P(n, r) definition moved to the start of the stem ('Here $P(n, r)$ denotes ...') in U04-A-004, 012, 017, 035 and the rewritten 036; stems now end with 'equals'/'is'. Options unchanged. | U04-A-004; U04-A-012; U04-A-017; U04-A-035 |
| ERR-078 | U04-C-003 JEE-MATH-U04-110 | E7 | minor | All 53 batch-C records (U04-108 to U04-160) have topic = 'Permutations and combinations', the chapter name, whereas batches A and B use real topics. Topic-based filtering in the generator therefore cannot separate distribution/stars-and-bars, grouping and select-then-arrange items. | **fixed** Topics re-tagged: Selection and arrangement (12), Permutations with restrictions (6), Selections with restrictions (1), Division into groups (10), Distribution and integer solutions (24). | - |
| ERR-079 | U05-B-030 JEE-MATH-U05-077 | E2 | critical | Stem says 'For a positive integer n'; at n = 1 distractor C ($2n$) also equals the true ratio $2$, so C is a correct value for one admissible n. | **fixed** Confirmed with sympy: at n = 1 distractor 2n equals the key 2. Stem now reads 'For every positive integer n'; distractor 2n replaced by (2n+1)/(n+1) (indices one more than given, mirroring the existing one-less slip (2n-1)/n). sympy: no option equals 2 and no two options coincide for any positive integer n; check extended to n = 1..10; rationales now show the computation. | U05-B-030 |
| ERR-080 | U05-B-011 JEE-MATH-U05-058 | E5 | minor | Distractor rationale for B says 'Rejects Statement I for the same reason', but no earlier rationale gives any reason for rejecting Statement I (A accepts it). So the rationale does not explain how B arises. | **fixed** Rationales made self-contained: the two Statement I rejections now state the slip (odd positions hold 7 coefficients against 6, so the odd sum is assumed larger); the TT rationale states Statement I was verified correctly. | U05-B-011 |
| ERR-081 | U05-A-040 JEE-MATH-U05-040 | E5 | minor | DR A reads 'Makes both of the above slips', but the slips are described in DR B and DR C, which are listed after it. | **fixed** Confirmed: the 'both slips' rationale referred to slips listed after it. Rewrote it to be self-contained: requiring only 5 \| r gives 5 rational terms (r = 0, 5, 10, 15, 20, verified), and using (n-r+1)/r for C(20,r+1)/C(20,r) gives 21 - r = 2r, r = 7; so both statements rejected. Key (both true) re-verified with sympy: rational r = 0, 10, 20 and C(20,7) = 2 C(20,6) only at r = 6. Stem, options and key unchanged. | - |
| ERR-082 | U05-B-035 JEE-MATH-U05-082 | E5 | minor | DR C reads 'Makes both of the above errors', but the second error (DR D) is listed below it. | **fixed** FF rationale rewritten to name both slips explicitly ((2^5)^2 = 1024 for the sum of squares; even-index sum assumed larger because it starts with C(9,0)) instead of 'both of the above errors'. | - |

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| ERR-083 | U05-C-012 JEE-MATH-U05-106 | E5 | minor | DR A ('Assumes the expression is always even') does not give 14 by itself; it also needs the factor 7. The assumption is also false at n = 2, where the value is 49, which is odd. | **fixed** Confirmed with sympy: n = 2 gives 49 (odd), so 'always even' alone does not yield 14. Rationale for the 14 distractor rewritten: sees the multiple of 7, wrongly believes the expression is always even, gives 2*7 = 14. Stem, options and key unchanged. | - |
| ERR-084 | U05-B-035 JEE-MATH-U05-082 | E6 | minor | Both statements repeat other items in the slice with trivially changed numbers, and no family_dependencies entry links them. Statement I (sum of squares of C(5,r) = C(10,5)) is U05-055 (n = 8). Statement II (even-index sum = odd-index sum for n = 9) is the fact tested in U05-052 and in U05-058 Statement I. | **aliased** Related tasks with different data and answers, not duplicates. Alias pairs requested with U05-B-008 (sum of squares), U05-B-005 and U05-B-011 (odd/even equal sums). | U05-B-011 |
| ERR-085 | U05-B-011 JEE-MATH-U05-058 | E6 | minor | Statement II (position of the middle term for n = 12) is the task of U05-056 (n = 15) with trivially changed numbers. Statement I repeats U05-052. Neither overlap is linked. | **aliased** Related tasks with different data; alias pairs requested with U05-B-005 (odd/even sums) and U05-B-009 (middle-term position). | U05-B-011 |
| ERR-086 | U05-A-040 JEE-MATH-U05-040 | E6 | minor | Statement I (count the rational terms of $(\sqrt2 + \sqrt[5]{3})^{20}$, answer 3) is the same task as U05-009 ($(\sqrt2+\sqrt[3]{3})^{12}$, answer 3) and U05-019 with trivially different numbers. They are not linked. | **aliased** Statement I counts rational terms of (sqrt2 + 3^(1/5))^20 (answer 3), the same method as U05-A-009 (answer 3) and U05-A-019 with different data; it is one half of a statement item, so a related task, not a duplicate. Alias group requested; no rewrite. | - |
| ERR-087 | U05-A-010 JEE-MATH-U05-010 | E6 | minor | Statement I (whether $(x+1/x)^{11}$ has a term independent of x) is the existence task of U05-013 and U05-046 (family U05-A-indep-term-existence) and is not linked. | **aliased** Statement I (no constant term in (x + 1/x)^11, powers 11 - 2r always odd, verified) is the existence task of U05-A-013 and U05-A-046 with different data, inside a two-statement item. Related task; alias group requested; no rewrite. | - |
| ERR-088 | U05-B-011 JEE-MATH-U05-058 | E9 | minor | The stem has ordinal numbers outside math mode ('1st, 3rd, 5th, ...', '6th term'). Other stems in the unit write '$5$th', '$4$th'. | **fixed** Ordinals written as $1$st, $3$rd, $5$th, $\ldots$ and $6$th in stem, rationales and explanation. | U05-B-011 |

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| ERR-089 | U05-B-002 JEE-MATH-U05-049 | E9 | minor | Several explanations write numbers outside math mode while the rest of the bank uses $...$. Affected: 049 ('there are 7 terms'), 060 ('9 terms'), 067 ('degree at most 7'), 071 ('With 8 terms'), 072 ('9 terms'), 073 ('7 terms'), 080 ('5 of them'), 089 ('the 101 powers'), 093 ('summing to 22'). | **fixed** Wrapped all listed numbers in $...$ in explanations (049, 060, 067, 071, 072, 073, 080, 089, 093); also swept rationales, misconceptions and difficulty reasons batch-wide (004, 006, 007, 010, 012, 016, 017, 021, 025, 026, 027, 028, 033, 039, 040, 042, 043, 047 local items). Regex scan now finds no digits outside math mode. | - |
| ERR-090 | U06-A-011 JEE-MATH-U06-011 | E6 | major | Items 004, 011 and 017 are the same task: the sum of the first n terms of an AP given a, d and n. 011 and 017 only add a short context wrapper. They sit in three different families (U06-A-sum-basic, U06-A-word-savings, U06-A-word-seating) with no shared family_dependencies entry, so one generated paper can contain all three. | **rewritten** Confirmed with sympy: old 011 (a=500, d=100, S_12=12600) was the same S_n-from-a,d,n task as 004 (S_20=860) with a context wrapper. Rewritten with new data and a different unknown: savings a=300, d=150; find n such that the nth month's saving equals the total of the first 3 months (S_3=1350, n=8). New family U06-A-word-term-equals-sum; qtype Multi-step; numerical alternate added (8). Distractors 7 (a+nd), 17 (half dropped in S_3), 5 (S_3 taken as 3a), each verified. | U06-A-011 |
| ERR-091 | U06-A-017 JEE-MATH-U06-017 | E6 | major | Same task as 004 (and 011): S_n of an AP from a, d and n, with a context wrapper. It has no shared family_dependencies entry with them. | **rewritten** Confirmed: old 017 (a=20, d=2, n=25, S=1100) repeated 004's task. Rewritten to find the first term from the total: 16 rows, d=2, 800 seats gives front row a=35. New family U06-A-word-first-term-from-sum; numerical alternate added (35). Distractors 34 (nd for (n-1)d), 10 (half dropped), 50 (S/n average row), each verified. Checked it does not duplicate 016 (finds d from a, l, S) or 038 (last term from sum). | U06-A-017 |
| ERR-092 | U06-C-041 JEE-MATH-U06-161 | E6 | minor | After the substitution this is the same minimisation as 125: t + 4/t >= 4 with t > 0, and the answer is 4 in both. The families (U06-C-amgm-logs and U06-C-amgm-exponential) are not linked. | **rewritten** Confirmed: after substitution both items minimise t + 4/t (t > 0) with key 4. Rewrote U06-C-041 as a reverse (parameter) item: least value of log_3 x + log_x k over x > 1 is 6, find k (key 3^9); new numbers, different unknown, new family U06-C-amgm-logs-param; question_type now Inverse/parameter MCQ, still Medium. | U06-C-041 |
| ERR-093 | U06-A-046 JEE-MATH-U06-046 | E6 | minor | The second half of 046 is exactly the task of 022: the greatest partial sum of a decreasing AP, found by locating the last positive term. Its family_dependencies does not list U06-A-max-partial-sum, so both can appear in one paper and test that skill twice. | **aliased** Confirmed: 046 finds d=-4 from S_10=S_15, then maximises S_n (S_12=S_13=312), whose second half is the greatest-partial-sum skill of 022 (S_14=287). Related task with different data and an extra first step, so not rewritten; alias requested so 022 and 046 never appear in one paper. | - |
| ERR-094 | U06-B-047 JEE-MATH-U06-107 | E6 | minor | The first stage of 107 is the task of 080: find r from the sum and the sum of squares of an infinite GP, by dividing to get (1-r)/(1+r) or a/(1+r). The families (U06-B-tail-estimate and U06-B-sum-and-squares) are not linked. | **aliased** Confirmed with sympy: 107 (S = 4/3, sum of squares 16/3) gives a = 2, r = -1/2 by the same elimination as 080 (r = 1/2). Different data, different answer and 107 goes on to a tail estimate, so it is a related task, not a duplicate. Alias requested for U06-B-047 with U06-B-020; no text changed. | - |

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| ERR-095 | U06-C-026 JEE-MATH-U06-146 | E7 | minor | The topic is 'Mixed AP and GP problems', but the item has no AP in it. It is a GP-and-logarithm item of the same kind as 093, which is filed under 'Geometric progression'. | **fixed** Confirmed no AP condition is used. Topic changed to 'Geometric progression' (subtopic 'Exponents in GP, bases unknown' kept), matching U06-B-033. | - |
| ERR-096 | U06-C-050 JEE-MATH-U06-170 | E7 | minor | The topic is 'Mixed AP and GP problems', but the item uses only a GP and the AM-GM bound r + 1/r. There is no AP. | **fixed** Confirmed only a GP and r + 1/r bounds are used. Topic changed to 'Relation between AM and GM'. | - |
| ERR-097 | U06-B-021 JEE-MATH-U06-081 | E7 | minor | Graded Medium, although it is a single subtraction T_6 = S_6 - S_5. It is also labelled 'Single-step computation MCQ', and its AP twin 005 (a_10 = S_10 - S_9) is graded Easy. | **rewritten** Confirmed: T_6 = S_6 - S_5 = 160 is one-step work, not Medium. Instead of relabelling, the item was made genuinely Medium to keep quotas: same S_n = 5(2^n - 1), now asks for the sum of the squares of the first 5 terms (recover T_n = 5*2^(n-1), squares form a GP with a = 25, r = 4, sum 8525). New distractors 34125 (six terms), 1705 (squares only the ratio), 24025 (S_5 squared), all verified. question_type changed to Multi-step computation MCQ; difficulty_reason, explanation, check and numerical alternate (8525) updated. | U06-B-021 |
| ERR-098 | U06-B-054 JEE-MATH-U06-114 | E5 | minor | The key is the only option written with set difference, (0,10) \setminus {5}. The closest distractor B is written as a union, (0,5) ∪ (5,10]. The odd form marks the key and makes the comparison with B harder than it needs to be. | **fixed** Key rewritten as $(0, 5) \cup (5, 10)$, matching the form of the other interval options; explanation updated to the same form. Value (sympy Union) unchanged. | U06-B-054 |
| ERR-099 | U07-C-008 JEE-MATH-U07-116 | E6 | major | Near-duplicate of U07-090: both evaluate g(x) = [x] + [-x] at x = 2 (090 asks the two-sided limit, 116 asks both one-sided limits and g(2)). 116's answer contains 090's answer outright. Families differ (U07-B-gif-one-sided vs U07-C-gif-sum-neg), so the paper generator can put both on one paper. The same fact also appears in U07-049 Statement I and U07-148 Statement I ([x] + [-x] at x = 1). | **rewritten** Confirmed: 116 evaluated [x] + [-x] at x = 2, which gives U07-090's answer outright and repeats the fact used in U07-049 and U07-148. Rewrote U07-C-008 with a different function and point: g(x) = x + [-x] at x = 3, asking lim_{3^-} g, lim_{3^+} g and g(3). Key is now 0, -1, 0, re-verified by one-sided numeric evaluation. Distractors: 1, 0, 0 from [-x] = -[x]; 0, 0, 0 from substituting x = 3 in both limits; 0, -1, -1 from applying -[x] - 1 at the integer. New family U07-C-gif-x-plus-neg-gif. Still Easy. | U07-C-008; U07-C-040 |
| ERR-100 | U07-C-032 JEE-MATH-U07-140 | E6 | major | Near-duplicate of U07-102: both use the identical function f(x) = lim_{n->inf} (x^{2n} - 1)/(x^{2n} + 1). 102 asks whether lim_{x->1} f exists, which is the jump analysis 140 needs to count discontinuities (and the answer follows by symmetry). Not linked: 102 is family U07-B-limit-defined-fn and 140 is U07-C-limit-defined-fn. 140 is linked only to 157. | **rewritten** Confirmed: 140 used the same f(x) = lim (x^{2n} - 1)/(x^{2n} + 1) as U07-B-046. Rewrote U07-C-032 with f(x) = lim (x^{2n+1} + x^2)/(x^{2n} + 1), which is x^2 for \|x\| < 1 and x for \|x\| > 1, with f(1) = 1 and f(-1) = 0. f is continuous at 1 but jumps at -1, so the key is now 1 (re-checked with mpmath at n = 10^7). The new twist is the asymmetry between the two boundary points. Distractors are 2 (counts both boundary points), 0 | U07-C-032 |

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| | | | | | (wrong symmetry argument) and 3. New family U07-C-limit-defined-fn-odd; it does not share data with U07-C-049. Still Medium. | |
| ERR-101 | U07-C-002 JEE-MATH-U07-110 | E6 | major | Same task as U07-059 with trivially different numbers: both reduce to lim_{x->0} sin(ax)/tan(bx) = a/b. The 'value for continuity' wrapper in 110 adds nothing, and the distractor patterns match (invert the ratio, multiply the coefficients, ignore the coefficients). Families differ (U07-B-std-trig vs U07-C-trig-removable). | **rewritten** Confirmed: sin 3x / tan 2x reduces to the same a/b task as U07-B-003. Rewrote U07-C-002 as f(x) = sin 2x / x + x sin(1/x) with f(0) = k. The sandwich bound \|x sin(1/x)\| <= \|x\| gives k = 2 (sympy limit). Distractors: 3 from misusing sin(theta)/theta with theta = 1/x -> infinity, 1 from dropping the coefficient, 1/2 from inverting it. New family U07-C-squeeze-removable. Still Easy. | U07-C-002 |
| ERR-102 | U07-C-001 JEE-MATH-U07-109 | E6 | major | Same task as U07-071 with trivially different numbers: both are lim_{x->a} (x^n - a^n)/(x - a) = n a^{n-1} (071: n = 5, a = 2; 109: n = 3, a = 3, framed as 'k for continuity'). Families differ (U07-B-xn-an vs U07-C-removable-rational-value). | **rewritten** Confirmed: (x^3 - 27)/(x - 3) is the x^n - a^n task of U07-B-015. Rewrote U07-C-001 as (x^3 - x^2 - 8x + 12)/(x - 2)^2 with f(2) = k. The numerator is (x - 2)^2 (x + 3), a double factor, so k = 5 (sympy limit). Distractors: 0 from cancelling only once, 2 from keeping leading terms only, -1 from a sign slip in the cofactor. Uses a different cubic from U07-C-035. New family U07-C-removable-double-factor. The numerical alternate now has answer 5. Still Easy. | U07-C-001 |
| ERR-103 | U07-B-037 JEE-MATH-U07-093 | E6 | major | U07-060 gives away the answer of U07-093. 060 shows lim (1 - cos 4x)/x^2 = 8. Since x sin x ~ x^2, 093 ('lim (1 - cos kx)/(x sin x) = 8, find k') is the same computation run backwards with the same numbers, so k = 4. Families differ (U07-B-one-minus-cos vs U07-B-param-trig). | **rewritten** Confirmed with sympy: (1 - cos kx)/(x sin x) and (1 - cos kx)/x^2 share the limit k^2/2, so with value 8 the item is U07-B-004 run backwards (k = 4). Rewrote U07-B-037 as: lim (1 - cos x cos 2x)/sin^2 kx = 1/2, k nonzero, find k^2 (key 5). New numbers, parameter moved to the denominator, product-of-cosines twist; distractors 10, 3, 9 each verified from the named slip. New family U07-B-param-cos-product; still Medium, Inverse/parameter MCQ, numerical alternate answer 5. | U07-B-037 |
| ERR-104 | U07-C-030 JEE-MATH-U07-138 | E6 | minor | The left-hand piece of U07-138, (1 - cos 4x)/x^2 as x -> 0^-, is exactly the item U07-060 (value 8), and 8 is 138's key. A student who has seen 060 gets half of 138 for free. Not linked (U07-C-two-sided-match vs U07-B-one-minus-cos). | **fixed** Confirmed: the left piece was exactly U07-B-004. Changed it to (1 - cos 8x)/(4x^2), whose limit is still 8, so it matches the unchanged right piece and the key stays 8 (sympy). The 16 distractor (no 1/2 factor: 64/4) still holds. The k-not-squared distractor is now 1 instead of 2. Explanation and numerical alternate updated. The item no longer contains U07-060, so no alias is needed. | U07-C-030 |
| ERR-105 | U07-C-016 JEE-MATH-U07-124 | E6 | minor | Near-duplicate of U07-065. Both factor a quadratic that contains (x - 3) over \|x - 3\| and resolve the modulus on the left of 3. 124 also asks for the right-hand limit, and its left-hand value comes from the same step as 065. Not linked (U07-B-modulus-one-sided vs U07-C-jump-size). The same pattern recurs in U07-148 Statement II, (x^2 - 1)/\|x - 1\|. | **aliased** Related, not a duplicate. 124 asks for the jump size of (x^2 - 9)/\|x - 3\| at 3 (key 12, needs both one-sided limits). 065 asks for one left-hand limit of a different quadratic (key -4). Same modulus-resolution mechanism, different data and answers, so I requested an alias with U07-B-009. The U07-148 overlap is gone because U07-C-040 was rewritten. | U07-C-040 |
| ERR-106 | U07-A-040 JEE-MATH-U07-040 | E6 | minor | U07-040 and U07-031 are the same task (fundamental period of a sum of two sinusoids as the LCM of their periods) with different frequencies. They are not linked (U07-A-per-sum-sinusoids vs U07-A-per-rational-frequencies), and both have an 'HCF of the periods' or 'period of one term' distractor. | **aliased** Confirmed related, not duplicate: 040 (sin(pi x/2) + cos(pi x/3), periods 4 and 6, period 12) and 031 (sin 3x + cos 2x, periods 2pi/3 and pi, period 2pi) use the same LCM-of-periods method with different data and different answers (both re-verified by the min_period check). Alias group requested; no rewrite. | - |

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| ERR-107 | U07-A-034 JEE-MATH-U07-034 | E6 | minor | Cross-half overlap: U07-034 (f(x) = x^3 + x + 1, find f^{-1}(3) = 1) supplies the key intermediate step of U07-195 (g = f^{-1} for the same f, find g'(3) = 1/f'(1) = 1/4) and of U07-212 (g''(3)). Not linked (U07-A-inv-value vs U07-D-inverse-fn-derivative). | **fixed** Confirmed: 034 used f(x) = x^3 + x + 1 with f^{-1}(3) = 1, the key intermediate of U07-D-030 (g'(3)) and U07-D-047 (g''(3)). Changed the function to f(x) = x^3 + 2x - 1 and asked f^{-1}(2): x^3 + 2x - 3 = (x - 1)(x^2 + x + 3), discriminant -11, so the key stays 1 (sympy solveset gives {1}). Distractors updated to 11 (f(2)), 1/11 (1/f(2)) and cbrt(2) (cubic term only); explanation and numerical alternate updated. The function no longer coincides with the U07-D items, so no alias is needed. | U07-A-034 |
| ERR-108 | U07-A-051 JEE-MATH-U07-051 | E3 | minor | The explanation rules out n <= -4 with the wrong inequality. It says 'sqrt(3 - n) < -n fails the bound', but sqrt(3 - n) < -n gives x = -sqrt(3 - n) > n, which is the lower bound being satisfied. The candidate fails the upper bound x < n + 1, because sqrt(3 - n) <= -(n + 1), i.e. n^2 + 3n - 2 >= 0, for n <= -4. | **fixed** Confirmed: for n <= -4 the candidate x = -sqrt(3 - n) fails the upper bound x < n + 1, not the lower bound. Explanation now states (n + 1)^2 - (3 - n) = n^2 + 3n - 2 > 0, so x > n + 1 and lies outside [n, n + 1) (checked numerically for n = -4 to -12). Key 2 unchanged, check passes. | - |
| ERR-109 | U07-A-032 JEE-MATH-U07-032 | E7 | minor | source_class is XII, but the modulus, signum and greatest integer functions and their graphs are in NCERT Class XI, Chapter 2 (Relations and Functions, 'Some functions and their graphs'). The XII tag comes from the U07-A batch brief, not from NCERT placement. | **fixed** Modulus/greatest-integer/signum graph item; source_class XII -> XI per NCERT Class XI placement. | - |
| ERR-110 | U07-A-036 JEE-MATH-U07-036 | E7 | minor | source_class is XII, but the modulus, signum and greatest integer functions and their graphs are in NCERT Class XI, Chapter 2 (Relations and Functions, 'Some functions and their graphs'). The XII tag comes from the U07-A batch brief, not from NCERT placement. | **fixed** Modulus graph item; source_class XII -> XI. | - |
| ERR-111 | U07-A-037 JEE-MATH-U07-037 | E7 | minor | source_class is XII, but the modulus, signum and greatest integer functions and their graphs are in NCERT Class XI, Chapter 2 (Relations and Functions, 'Some functions and their graphs'). The XII tag comes from the U07-A batch brief, not from NCERT placement. | **fixed** Modulus graph item; source_class XII -> XI. | - |
| ERR-112 | U07-A-038 JEE-MATH-U07-038 | E7 | minor | source_class is XII, but the modulus, signum and greatest integer functions and their graphs are in NCERT Class XI, Chapter 2 (Relations and Functions, 'Some functions and their graphs'). The XII tag comes from the U07-A batch brief, not from NCERT placement. | **fixed** Greatest integer item; source_class XII -> XI. | - |
| ERR-113 | U07-A-043 JEE- | E7 | minor | source_class is XII, but the modulus, signum and greatest integer functions and their graphs are in NCERT Class XI, Chapter 2 (Relations and | **fixed** Modulus graph item; source_class XII -> XI. | - |

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| | MATH-U07-043 | | | Functions, 'Some functions and their graphs'). The XII tag comes from the U07-A batch brief, not from NCERT placement. | | |
| ERR-114 | U07-A-044 JEE-MATH-U07-044 | E7 | minor | source_class is XII, but the modulus, signum and greatest integer functions and their graphs are in NCERT Class XI, Chapter 2 (Relations and Functions, 'Some functions and their graphs'). The XII tag comes from the U07-A batch brief, not from NCERT placement. | **fixed** Signum item; source_class XII -> XI. | - |
| ERR-115 | U07-A-049 JEE-MATH-U07-049 | E7 | minor | source_class is XII, but the modulus, signum and greatest integer functions and their graphs are in NCERT Class XI, Chapter 2 (Relations and Functions, 'Some functions and their graphs'). The XII tag comes from the U07-A batch brief, not from NCERT placement. | **fixed** Greatest integer and signum statements; source_class XII -> XI. | - |
| ERR-116 | U07-A-051 JEE-MATH-U07-051 | E7 | minor | source_class is XII, but the modulus, signum and greatest integer functions and their graphs are in NCERT Class XI, Chapter 2 (Relations and Functions, 'Some functions and their graphs'). The XII tag comes from the U07-A batch brief, not from NCERT placement. | **fixed** Greatest integer equation; source_class XII -> XI. | - |
| ERR-117 | U07-A-053 JEE-MATH-U07-053 | E7 | minor | source_class is XII, but the modulus, signum and greatest integer functions and their graphs are in NCERT Class XI, Chapter 2 (Relations and Functions, 'Some functions and their graphs'). The XII tag comes from the U07-A batch brief, not from NCERT placement. | **fixed** Signum and greatest integer item; source_class XII -> XI. | - |
| ERR-118 | U07-A-054 JEE-MATH-U07-054 | E7 | minor | source_class is XII, but the modulus, signum and greatest integer functions and their graphs are in NCERT Class XI, Chapter 2 (Relations and Functions, 'Some functions and their graphs'). The XII tag comes from the U07-A batch brief, not from NCERT placement. | **fixed** Greatest integer item; source_class XII -> XI. Batch class split is now 38 XI / 18 XII (was 28/28 per the original brief); difficulty quotas unchanged. | - |
| ERR-119 | U07-E-041 JEE-MATH-U07-263 | E6 | major | Near-duplicate of JEE-MATH-U07-241 with no shared family_dependencies entry. Both are the same task: liquid or sand fills a cone whose radius is a fixed multiple of its height, dV/dt is given, find dh/dt at a given depth. | **rewritten** Confirmed: 263 (r = 6h, dV/dt = 12, h = 4, dh/dt = 1/(48 pi)) repeated the task of 241 (find dh/dt in a cone with fixed r:h ratio). Rewritten to a different unknown: sand pile with h = r/3, dV/dt = 12 cm^3/s, find the rate of increase of the BASE AREA when h = 3 (r = 9): V = pi r^3/9, dr/dt = 4/(9 pi), dA/dt = 2 pi r dr/dt = 8 cm^2/s (sympy check). Distractors 8/3 (missing 1/3 in V), 24 (missing factor 3 in d(r^3)), 4 (dA/dt taken as pi r | U07-E-041 |

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| | | | | | dr/dt), each re-derived. New family U07-E-rate-cone-base-area; numerical alternate (8) added. Difficulty Medium unchanged. | |
| ERR-120 | U07-C-040 JEE-MATH-U07-148 | E6 | major | Statement I of 148 tests the same function and phenomenon as JEE-MATH-U07-116 ([x] + [-x] at an integer), and Statement II tests the same thing as JEE-MATH-U07-124 ((x^2 - a^2)/\|x - a\| has unequal one-sided limits at a). No family link exists, so 116 or 124 gives away the truth value of a statement in 148 if they appear together. | **rewritten** Confirmed: Statement I repeated [x] + [-x] at an integer, and Statement II repeated the (x^2 - a^2)/\|x - a\| jump. Rewrote U07-C-040 with new statements on the order of composition. Statement I: [\|x\|] is continuous at 0 (true; it is constant 0 on (-1, 1)). Statement II: \|[x]\| is continuous at 0 (false; the left limit is 1 and g(0) = 0). Both were checked numerically with one-sided evaluation. The key is now 'Statement I is true but Statement II is false'. New family U07-C-stmt-gif-mod-order. Still Medium. | U07-C-040; U07-C-008 |
| ERR-121 | U07-E-013 JEE-MATH-U07-235 | E6 | minor | Near-duplicate of JEE-MATH-U07-224, unlinked. Both rest on the single relation dV/dt = 4 pi r^2 dr/dt for a sphere. One item goes forwards (dr/dt to dV/dt) and the other goes backwards. | **aliased** Confirmed related, not duplicate: 224 (dr/dt = 1/2, r = 4, dV/dt = 32 pi) and 235 (dV/dt = 18 pi, diameter 6, dr/dt = 1/2) use the same relation dV/dt = 4 pi r^2 dr/dt in opposite directions with different data. Alias group requested. | - |
| ERR-122 | U07-E-052 JEE-MATH-U07-274 | E6 | minor | Statement II of 274 asserts that x^{1/x} reaches its maximum at x = e. That is the key monotonicity fact needed for JEE-MATH-U07-261 (greatest of 2^{1/2}, 3^{1/3}, 5^{1/5}, 7^{1/7}), so 274 leaks the method and most of the answer. The two items are not linked. | **aliased** Confirmed: Statement II of 274 (max of x^{1/x} is e^{1/e}, via (1 - ln x)/x^2) states the monotonicity fact behind 261, though 261 still needs the comparison across x = e (2^{1/2} = 4^{1/4} vs 3^{1/3}). Related, not a copy; alias group requested so they never share a paper. | - |
| ERR-123 | U07-E-037 JEE-MATH-U07-259 | E6 | minor | Close to JEE-MATH-U07-244, unlinked. Both use the discriminant of f'(x) = 3x^2 + 2bx + c for a monic cubic with a parameter in the x^2 coefficient. 244 asks for a nonpositive discriminant (monotonic) and 259 for a positive one (two extrema), so each item's answer is the other's main distractor. | **aliased** Confirmed: 244 (x^3 + ax^2 + 3x + 5 increasing iff a in [-3, 3]) and 259 (x^3 + 3kx^2 + 12x + 1 has both extrema iff \|k\| > 2) use complementary discriminant conditions; each answer mirrors the other's distractor. Different data and answers, so alias rather than rewrite. | - |
| ERR-124 | U07-D-052 JEE-MATH-U07-217 | E6 | minor | Near-duplicate of JEE-MATH-U07-207, unlinked. Both ask for the derivative of sin^{-1}(multiple-angle expression) with respect to sin^{-1}x, using the substitution x = sin(theta) and a branch decision. 207's distractor B is exactly the 'pi minus' branch that 217 is about. | **aliased** Related task, not a duplicate: different expression (double-angle vs triple-angle), different interval and branch, different answer (-2 vs 3; both re-verified with sympy). Aliased U07-D-052 with U07-D-042 so they never share a paper. | - |
| ERR-125 | U07-E-049 JEE-MATH-U07-271 | E6 | minor | Uses the same function f(x) = x^3 - 3x on [0, k] as Statement II of JEE-MATH-U07-252 (absolute maximum of x^3 - 3x on [0, 2] is at x = 2). If both appear together, 252 tells the student that f(2) > f(0), which rules out option D (0, 2] of 271. The items are not linked. | **fixed** Confirmed: Statement II of 252 (U07-E-030) used h(x) = x^3 - 3x on [0, 2], the function of 271, revealing f(2) > f(0). Changed 252 Statement II to h(x) = x^3 - 12x on [0, 4]: stationary point x = 2 with h(2) = -16, h(0) = 0, h(4) = 16, so the maximum is at x = 4 and the statement stays true (key Both true unchanged, sympy check updated). Rationales and explanation of 252 updated. 271 untouched; no alias needed. | U07-E-030 |

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| ERR-126 | U07-D-001 JEE-MATH-U07-166 | E7 | minor | source_class is XII, but differentiating a product of two polynomials by the product rule is NCERT Class XI material (Limits and Derivatives, algebra of derivatives). The bank already tags Limits items in this unit as XI. | **fixed** Product rule on polynomials is NCERT Class XI; source_class set to XI. Key 66 re-verified. | - |
| ERR-127 | U07-D-002 JEE-MATH-U07-167 | E7 | minor | source_class is XII, but the quotient rule on a rational function is NCERT Class XI (Limits and Derivatives). | **fixed** Quotient rule on a rational function is Class XI; source_class set to XI. Key -5/4 re-verified. | - |
| ERR-128 | U07-D-011 JEE-MATH-U07-176 | E7 | minor | source_class is XII, but the quotient rule with sin x and cos x is NCERT Class XI (Limits and Derivatives covers derivatives of trigonometric functions and the quotient rule). | **fixed** Quotient rule with sin and cos is Class XI; source_class set to XI. Key 1/(1+cos x) re-verified. | - |
| ERR-129 | U07-D-015 JEE-MATH-U07-180 | E7 | minor | source_class is XII, but the product rule on a product of linear polynomials is NCERT Class XI (Limits and Derivatives). | **fixed** Product rule on linear factors is Class XI; source_class set to XI. Key 11 re-verified. | - |
| ERR-130 | U07-D-044 JEE-MATH-U07-209 | E5 | minor | The rationale for distractor B refers to 'A + B' and 'dA = -dB', but the explanation names the substituted angles P and Q (x^3 = sin P, y^3 = sin Q). The letters A and B are not defined anywhere in the record, and they clash with the option letters. | **fixed** Rationale reworded to use the explanation's angles: P + Q constant, so dP = -dQ. | - |
| ERR-131 | U07-D-024 JEE-MATH-U07-189 | E5 | minor | The rationale for distractor A depends on another option's rationale ('as in the previous slip'), so it cannot be read on its own and breaks if options are reordered. | **fixed** Rationale made self-contained: y'' = e^x cos x gives 1 + k = 0 at x = 0, solved as k = 1 by a sign slip. Key -2 re-verified. | - |

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| ERR-132 | U07-C-053 JEE-MATH-U07-161 | E5 | minor | The rationale for distractor D says '[x]^2 - [x] = 0, i.e. at x = 0 and x = 1'. In fact [x]^2 - [x] = 0 on the whole interval [0, 2), not only at two points. The intended slip is about the integers m with m^2 - m = 0. | **fixed** Confirmed: [x]^2 - [x] = 0 on all of [0, 2), not only at two points. Reworded the rationale for 2: 'Thinks f is continuous at every integer m where m^2 - m = 0, i.e. m = 0 and m = 1, leaving only -1 and 2.' Options and key 3 are unchanged. | - |
| ERR-133 | U08-C-012 JEE-MATH-U08-124 | E6 | major | Near-duplicate of U08-027 (and of U08-143 in the second half of the unit): all three are the same task, the integral of x^(n-1)/(x^(2n)+a^2) by t = x^n to an inverse tangent, with trivially different numbers. Families differ (U08-C-sub-to-standard, U08-A-subst-to-arctan, U08-C-sub-power-standard) and none lists the others in family_dependencies, so the paper generator can place them together. | **rewritten** Confirmed with sympy: U08-C-012 (x/(x^4+1)), U08-A-027 (x^3/(1+x^8)) and U08-C-031 (x^2/(x^6+16)) are the same t = x^n to inverse tangent task. Rewrote U08-C-012 (still Easy, Single-step) as a logarithmic substitution into the 1/sqrt(u^2+a^2) log form: int dx/(x sqrt((log x)^2+4)) = log\|log x + sqrt((log x)^2+4)\| + C; new distractors, new family U08-C-sub-log-sqrt-standard. The remaining pair U08-C-031 / U08-A-027 is aliased. | U08-C-012 |
| ERR-134 | U08-C-027 JEE-MATH-U08-139 | E6 | major | Near-duplicate of U08-123 (same half) and of Statement II of U08-150 (second half): the integral of 1/sqrt((x-p)(q-x)) is completed to 1/sqrt(r^2-(x-m)^2) giving an inverse sine, with only the roots changed. Not linked by family_dependencies (U08-C-sqrt-product-form vs U08-C-sqrt-quad-complete-square vs U08-C-statements-b). | **rewritten** Confirmed: (x-1)(3-x) = 1-(x-2)^2 repeats U08-C-011 and Statement II of U08-C-038 with only the roots changed. Rewrote U08-C-027 (still Medium) as an Inverse/parameter item with a different answer form: int dx/sqrt((x-2)(8-x)) = k sin^-1 sqrt((x-2)/m) + C, k, m > 0; key k + m = 8 (k = 2, m = 6, verified by sympy coefficient matching); numerical alternate added; new family U08-C-sqrt-product-root-form. U08-C-011 / U08-C-038 aliased under U08-150. | U08-C-027 |
| ERR-135 | U08-B-035 JEE-MATH-U08-091 | E6 | major | Near-duplicate of U08-066: both are the integral of 1/(x(x^n+1)), n = 4 and n = 2. The 'multiply by x^(n-1), put t = x^n' method of U08-091 solves U08-066 identically, and the answers have the same shape (1/n)log(x^n/(x^n+1)). Families U08-B-pf-x-xn and U08-B-pf-quadratic-factor are not linked. | **rewritten** Confirmed: 1/(x(x^4+1)) and U08-B-010's 1/(x(x^2+1)) are the same task (1/(x(x^n+1))) with the same answer shape. Rewrote U08-B-035 as the integral of x^3/((x^2+1)(x^2+3)): substitute t = x^2 for the odd integrand, then split t/((t+1)(t+3)) = -1/(2(t+1)) + 3/(2(t+3)); key (3/4)log(x^2+3) - (1/4)log(x^2+1) (sympy apart and derivative check). New distractors from dropping the 1/2 of x dx = dt/2, swapping the cover-up coefficients, and a numerator sign slip at t = -1. New family U08-B-pf-odd-power-sub; still Medium. Not of the form 1/(x(x^n+1)), so no alias needed. | U08-B-035 |
| ERR-136 | U08-B-036 JEE-MATH-U08-092 | E6 | major | Near-duplicate of U08-037: the integral of 1/(e^x - 1) versus 1/(1 + e^x), same task with one sign changed. Families U08-B-pf-exp-sub and U08-A-subst-exp-divide are not linked, so both can appear in one paper and the second is solved by imitating the first. | **rewritten** Confirmed: 1/(e^x - 1) is U08-A-037's 1/(1 + e^x) with one sign changed, solvable by the same add-and-subtract trick. The suggested quadratic in e^x is already used by U08-B-054, so U08-B-036 was rewritten as the integral of (log x)^2/(x(log x + 1)(log x + 2)): t = log x, then the improper fraction needs division, t^2/((t+1)(t+2)) = 1 + 1/(t+1) - 4/(t+2); key log x + log\|log x + 1\| - 4log\|log x + 2\| (sympy verified). Distractors: skipping the division (loses the log x term), a remainder sign slip, back-substituting t as x. New family U08-B-pf-log-sub-improper; still Medium. | U08-B-036 |
| ERR-137 | U08-B-042 JEE-MATH- | E6 | minor | After its intended substitution x = e^t, U08-098 is exactly U08-061 (the integral of e^t(sin t + cos t)). Families U08-B-parts-logs-ln-sub and U08-B-ex-f-fprime-trig are not linked; one item makes the other a one-step exercise. | **aliased** Confirmed related: after x = e^t, U08-B-042 reduces to U08-B-005's integral of e^t(sin t + cos t). The extra substitution step makes it a related task rather than a copy, so per | - |

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| | U08-098 | | | | the minor-severity rule the two are aliased (never in the same paper) instead of rewritten. | |
| ERR-138 | U08-A-047 JEE-MATH-U08-047 | E6 | minor | U08-047 reduces in one identity step to the integrand of U08-013 minus 3, so U08-013's answer gives away most of U08-047's key. Families U08-A-identity-cancel and U08-A-pythag-split are not linked. | **aliased** Confirmed related, not duplicate: sympy confirms sin^6 x + cos^6 x = 1 - 3 sin^2 x cos^2 x, so 047's integrand is 013's integrand minus 3 and 047's key tan x - cot x - 3x builds on 013's key tan x - cot x. Different integrand, different answer, and 047 adds the sixth-power identity step, so no rewrite. Alias group requested so the two never share a paper. | - |
| ERR-139 | U08-A-042 JEE-MATH-U08-042 | E6 | minor | Statement I of U08-042 (the integral of cos^2 x, with the sin^2 x sign) is the mirror of U08-004 (the integral of sin^2 x), whose distractor B is exactly the correct form for cos^2 x. One item settles the other. Families U08-A-stmt-standard-formulas and U08-A-half-angle-square are not linked. | **aliased** Confirmed related, not duplicate: sympy confirms d/dx(x/2 + sin2x/4) = cos^2 x, so Statement I of 042 (false, sin^2 x form) mirrors 004 (int sin^2 x = x/2 - sin2x/4) and one item hints at the other. 042 also tests int 3^{2x} dx (9^x/log 9, not 9^x/log 3) and asks a different task, so no rewrite. Alias group requested. | - |
| ERR-140 | U08-B-041 JEE-MATH-U08-097 | E6 | minor | Statement II of U08-097 (the integral of x cos^2 x) is decided directly by U08-058's result (the integral of x cos 2x). The explanation of U08-097 quotes U08-058's answer verbatim. Families U08-B-statements-division-parts and U08-B-parts-poly-trig are not linked. | **aliased** Confirmed: Statement II of U08-B-041 (integral of x cos^2 x) uses U08-B-002's result for the integral of x cos 2x. It is a related task, not a duplicate, so the two are aliased. | - |
| ERR-141 | U08-C-016 JEE-MATH-U08-128 | E6 | minor | Statement II of U08-128 (the integral of 1/(9-x^2) with the ratio inverted) is the same task as U08-116 (the integral of 1/(16-x^2)), and the false claim is U08-116's distractor A pattern. Statement I of U08-128 is the same task as U08-113. Not linked by family_dependencies (U08-C-statements-a vs U08-C-a2-minus-x2 / U08-C-x2-plus-a2). The same pattern recurs between U08-120 and Statement I of U08-150 (sqrt(x^2+a^2) with the sign of the sqrt(x^2-a^2) formula). | **aliased** Confirmed: Statement I of U08-C-016 is the task of U08-C-001 and Statement II is the U08-C-004 task with its ratio-inversion distractor. Related statement-check items, not copies: alias links requested (U08-C-016 with U08-C-001 and with U08-C-004). The U08-C-008 / U08-C-038 pattern mentioned here is aliased under U08-150. | - |
| ERR-142 | U08-A-002 JEE-MATH-U08-002 | E9 | minor | Logarithm convention: the first half of unit 8 writes bare \log in answers where the base matters (it must be natural log), while the second half of the same unit (U08-169 onward) writes \log_e. Students must assume log = ln, and keys such as -3^-x/log 3 are wrong if log is read as log10. | **retained** Log-notation finding; per errata instructions it is ignored here because the editor unifies \log notation bank-wide afterwards. No change to U08-A-002, 012, 017, 042, 050. | - |
| ERR-143 | U08-C-026 JEE-MATH-U08-138 | E5 | minor | The distractor rationale for option A has an unfinished placeholder and does not show how option A arises. | **fixed** Confirmed the placeholder rationale. Rewrote it: the slip treats x^2-4x+3 as (x-1)^2-a^2 with u = x-1 (shifting by the smaller root) and writes log\|u + sqrt(x^2-4x+3)\|, which is exactly the option log\|x-1+sqrt(x^2-4x+3)\|; noted that x^2-4x+3 = (x-1)^2-2(x-1) is not of that form. | - |

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| ERR-144 | U08-B-019 JEE-MATH-U08-075 | E5 | minor | The distractor rationale for option A opens with 'Same wrong identity', which points to the rationale for D that comes after it. Under that wrong identity the bracket is (1/2)cosec^2(x/2) + cot(x/2), and f = -cot(x/2) does not satisfy f + f' = bracket, so the rationale does not produce option A on its own terms. | **fixed** Confirmed: under 1 + cos x = 2sin^2(x/2) the bracket is (1/2)cosec^2(x/2) + cot(x/2), and f = -cot(x/2) gives f + f' = -cot(x/2) + (1/2)cosec^2(x/2), which is not that bracket (sympy). The rationale for -e^x cot(x/2) is now self-contained: it uses the wrong identity, integrates only the (1/2)cosec^2(x/2) piece to -cot(x/2), drops the cot(x/2) term and attaches e^x. Options and key unchanged. | - |
| ERR-145 | U08-C-035 JEE-MATH-U08-147 | E6 | major | U08-147 and U08-141 integrate the same quadratic x^2-4x+3 (U08-141 in u=e^x). The antiderivative that U08-147 needs and prints, (1/2)log\|(x-3)/(x-1)\|, becomes U08-141's key when x is replaced by e^x. The families U08-C-fvalue-log and U08-C-sub-exp-quadratic are not linked, so both can appear on one paper. | **rewritten** Confirmed: U08-C-035 prints (1/2)log\|(x-3)/(x-1)\|, which is U08-C-029's key with x for e^x. Rewrote U08-C-035 (still Medium) with a new quadratic and a different unknown: f:(-inf,2) -> R, f'(x) = 1/(x^2-6x+8), f(0) = 0, find lim_{x->-inf} f(x) = -(1/2)log 2 (sympy: definite integral from 0 to X, then limit). Distractors: constant taken as 0, inverted ratio, 1/a instead of 1/(2a). New family U08-C-fvalue-log-limit. | U08-C-035 |
| ERR-146 | U08-C-056 JEE-MATH-U08-168 | E6 | major | After u=x^2, U08-168 is (1/2) of int du/(u^2+u+1), and Statement II of U08-149 states that int dx/(x^2+x+1) = (2/sqrt3)tan^-1((2x+1)/sqrt3). Seeing U08-149 gives U08-168's key at once. The families U08-C-sub-power-quad and U08-C-statements-a are not linked. | **rewritten** Confirmed: after u = x^2, U08-C-056 is half of Statement II of U08-C-037. Rewrote U08-C-056 (still Difficult) as int x^3/(x^4+6x^2+10) dx: power substitution then a linear-over-quadratic split, key (1/4)log(x^4+6x^2+10) - (3/2)tan^-1(x^2+3) + C (sympy verified; quadratic u^2+6u+10 used nowhere else in the batch). New family U08-C-sub-power-linear-quad. | U08-C-056 |
| ERR-147 | U08-E-040 JEE-MATH-U08-264 | E6 | major | U08-264 asks for a in "area between y^2=4ax and x^2=4ay is 1/3". U08-241 is the same region with 4a=6 (area 12), and its explanation quotes the general result 16a^2/3. That one formula decides both items, and the families U08-E-param-two-parabolas and U08-E-two-parabolas are not linked. | **rewritten** U08-E-040 rewritten: parameter a from the area of {x^2 <= y <= ax^2, y <= 1} = 1/3 (key 16/9; horizontal strips between two coaxial parabolas). New family U08-E-param-nested-parabolas; no longer shares the 16a^2/3 result with U08-E-017. Distractors 4 (one half only), 4/3 (half-width sqrt(y)/a), 16 (area inside y = ax^2 alone) re-verified with sympy. | U08-E-040 |
| ERR-148 | U08-D-043 JEE-MATH-U08-211 | E6 | minor | After the parity step, U08-211 equals 4 x int_0^pi x sin x/(1+cos^2 x) dx, which is exactly U08-187 (key pi^2/4). U08-187 gives away U08-211's key (pi^2). The families U08-D-odd-even-king and U08-D-king-x-times are not linked. | **rewritten** Confirmed with sympy: after the parity step the old item was exactly 4 x U08-D-019 (int_0^pi x sin x/(1+cos^2 x) dx = pi^2/4), so it got its key directly from that item. Rewritten into a different chain on the same subtopic and difficulty: int_{-pi}^{pi} (x cos x + sin^4 x)/(sin^4 x + cos^4 x) dx = pi (odd part vanishes over an even denominator; f(2a - x) = f(x) doubling; King's property in ratio form). No x f(sin x) step remains. New family U08-D-odd-even-king-ratio; distractors pi/2, 2pi, pi/4 each from a named lost or extra doubling; check recomputes by numerical integration from the stem (matches pi to 30 digits). Difficult label kept. No alias needed. | U08-D-043 |
| ERR-149 | U08-C-052 JEE-MATH-U08-164 | E6 | minor | Statement II of U08-164 (coefficient of the inverse sine in int sqrt(1-4x^2) dx) is the same task as U08-151 (coefficient k in int sqrt(9-4x^2) dx), with a=1 in place of a=3. Statement I (int dx/(2x^2+2x+1)) is the same task as U08-136 in the first half (int dx/(2x^2+x+1)). None of these families are linked. | **aliased** Confirmed: Statement I of U08-C-052 is the U08-C-024 task and Statement II is the U08-C-039 coefficient task with a = 1. Related statement checks, not copies: alias links requested (U08-C-052 with U08-C-024 and with U08-C-039). | - |

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| ERR-150 | U08-C-038 JEE-MATH-U08-150 | E6 | minor | Statement I of U08-150 (the sign of the log term in int sqrt(x^2+1) dx) is decided by U08-120 in the first half (int sqrt(x^2+4) dx, whose key shows +2 log). The families U08-C-statements-b and U08-C-sqrt-x2-plus-a2-integrand are not linked. U08a already reports Statement II against U08-123 and U08-139. | **aliased** Confirmed: Statement I of U08-C-038 is decided by U08-C-008's key, and Statement II is the U08-C-011 task (2x-x^2 = 1-(x-1)^2). Alias links requested (U08-C-038 with U08-C-008 and with U08-C-011). U08-C-027 has been rewritten (U08-139), so it no longer needs a link. | U08-C-027 |
| ERR-151 | U08-C-053 JEE-MATH-U08-165 | E6 | minor | U08-165 integrates (x+5)/(3+2x-x^2). Its non-log part is 6 int dx/(3+2x-x^2), and U08-137 in the first half gives int dx/(3+2x-x^2) = (1/4)log\|(x+1)/(3-x)\| on the same interval (-1,3). U08-155 uses the same split-and-a^2-u^2 method on (x+1)/(5-4x-x^2). None of the families are linked (U08-C-fvalue-linear-quad, U08-C-a2-minus-quad, U08-C-linear-over-neg-quad). | **aliased** Confirmed: the non-log part of U08-C-053 is 6 times the U08-C-025 integral on the same interval, and U08-C-043 uses the same split method. U08-C-053 is a function-value item with a nonzero constant and two sign traps, so it is a related task, not a copy: alias links requested (U08-C-053 with U08-C-025 and with U08-C-043). | - |
| ERR-152 | U08-E-032 JEE-MATH-U08-256 | E6 | minor | Statement I of U08-256 (area enclosed by \|x\|+\|y\|=3) is the same task as U08-234 (area enclosed by \|x\|+\|y\|=2), with only the number changed. The families U08-E-stmt-standard and U08-E-mod-square are not linked. | **aliased** Verified: Statement I of U08-E-032 (\|x\|+\|y\|=3, true area 18) uses the same method as U08-E-010 (\|x\|+\|y\|=2, area 8) with different data and answer; related task, so alias requested rather than rewrite. | - |
| ERR-153 | U08-E-022 JEE-MATH-U08-246 | E6 | minor | U08-246 and U08-228 use the same ellipse x^2/16+y^2/9=1. U08-246's key is U08-228's key (the quarter ellipse, 3pi) minus the obvious triangle (6). The families U08-E-ellipse-chord and U08-E-ellipse-full are not linked. | **fixed** U08-E-022 moved to the ellipse x^2/25 + y^2/4 = 1 with chord x/5 + y/2 = 1; key 5pi/2 - 5, distractors 5pi/2, 15pi/2 + 5, 5pi - 5. No longer shares the ellipse or quarter-ellipse value with U08-E-004. | U08-E-022 |
| ERR-154 | U08-D-052 JEE-MATH-U08-220 | E6 | minor | Statement II of U08-220 (int_0^{pi/2} sin 2x log_e(tan x) dx = 0) uses exactly the King's-property sign flip of log_e(tan x) that is the whole content of U08-176 (int_0^{pi/2} log_e(tan x) dx = 0). The families U08-D-statement-fractional and U08-D-king-logtan are not linked. | **aliased** Confirmed related, not duplicate: Statement II of U08-D-052 (int_0^{pi/2} sin 2x log_e(tan x) dx = 0) uses the same sign flip of log_e(tan x) under x -> pi/2 - x as U08-D-008 (int_0^{pi/2} log_e(tan x) dx = 0), but it is one half of a two-statement item with a different integrand and a separate fractional-part statement (Statement I re-verified: integral = 5/2, so false; Statement II re-verified numerically = 0). Alias group requested; no rewrite. | - |
| ERR-155 | U08-E-031 JEE-MATH-U08-255 | E5 | minor | The distractor_review rationale for option A says "Makes both of the above slips", but it is listed first, before the B and C rationales it refers to. Read in order, it points to nothing. | **fixed** U08-E-031 'both false' rationale rewritten to be self-contained: reads semi-axes as 9 and 4 (36pi) and integrates sin x as cos x (-2). | - |

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| ERR-156 | U08-C-029 JEE-MATH-U08-141 | E9 | minor | Logarithm notation in this slice is inconsistent. U08-141 to U08-166 (batch C) write a bare \log, while U08-169 to U08-280 (batches D and E) write \log_e. The same paper can therefore mix log and log_e for the natural logarithm. This extends the U08a finding on U08-002 to the second half of the unit. | **retained** Log-notation finding; per the errata instructions the editor unifies \log notation bank-wide afterwards. No change in this batch. | - |
| ERR-157 | U08-E-005 JEE-MATH-U08-229 | E7 | minor | question_type "Graph/diagram MCQ" is applied to text-only area items that have no figure (figures = [], figure_required = false). Items of the same kind are tagged "Multi-step computation MCQ" (for example U08-247 and U08-260, which are also modulus-curve regions). The tag is therefore not informative for paper balancing. | **fixed** Graph/diagram MCQ removed from figure-less items: U08-E-005 and U08-E-014 now Single-step computation MCQ; U08-E-024, U08-E-041, U08-E-044, U08-E-048 now Multi-step computation MCQ. | - |
| ERR-158 | U09-A-018 JEE-MATH-U09-018 | E6 | major | Near-duplicate of U09-002 with no shared family_dependencies (U09-A-degree-fractional-power vs U09-A-degree-radical). Both use the same equation y'' = [1+(y')^2]^{3/2}, written with the sides swapped. U09-002 asks for the degree (2) and U09-018 asks for order and degree (2, 2), so either item gives away the other's answer. | **rewritten** Confirmed: same equation y'' = [1+(y')^2]^{3/2} as U09-A-002 with sides swapped; either key gives the other. The auditor's suggested equation would have duplicated U09-A-015 ((y'')^{2/3} = (1+y')^{1/2}), so rewrote 018 with a new twist: sqrt(1 + y'') = (y')^3 + y, where squaring leaves the highest derivative linear (order 2, degree 1, verified by sympy via the od() polynomial check). New distractors 2 and 2 / 2 and 6 / 1 and 6 from named slips; new family U09-A-degree-radical-on-highest. Still Medium, XII. | U09-A-018 |
| ERR-159 | U09-A-027 JEE-MATH-U09-027 | E6 | major | Uses the same DE as U09-019, dy/dx = (1+y^2)/(1+x^2), and the families are not linked. 019's key family y - x = C(1+xy) solves 027 in one line: y(0)=1 gives C=1, so y = (1+x)/(1-x) and y(1/2) = 3. | **rewritten** Confirmed with sympy: 019's family y - x = C(1+xy) with (0,1) gives C = 1 and y(1/2) = 3, 027's key. Rewrote 027 as (1+x^2) dy = 2(1+y^2) dx, y(0) = 1, domain 0 <= x < sqrt2 - 1, asking y(1/3): tan^{-1} y = 2 tan^{-1} x + pi/4, tan(2 tan^{-1}(1/3)) = 3/4, y = 7 (sympy). Distractors 2 (drops the 2), 5 (2 tan^{-1} x as tan^{-1} 2x), 1/7 (tan(pi/4 - theta)). New family U09-A-ps-tan-double-angle; numerical alternate updated to 7. | U09-A-027 |
| ERR-160 | U09-C-033 JEE-MATH-U09-127 | E6 | major | Same task as U09-025 with trivially different numbers, and the families are not linked (U09-C-app-cooling-k vs U09-A-app-cooling). Both use Newton cooling where the excess temperature halves in 10 minutes and the asked quantity sits at a quarter of the initial excess, 20 minutes in. | **rewritten** Confirmed: same Newton-cooling task as U09-A-025 (excess halves in 10 min, quarter point at 20 min). Rewrote U09-C-033 with a different unknown: room temperature S is unknown, readings 100, 70, 52 at t = 0, 10, 20 (excess ratio 3/5, not 1/2); key S = 25 from (70 - S)^2 = (100 - S)(52 - S). New family U09-C-app-cooling-ambient; question_type now Inverse/parameter MCQ; still Medium; numerical alternate answer 25; units as $^\circ\mathrm{C}$. | U09-C-033 |
| ERR-161 | U09-B-019 JEE-MATH-U09-066 | E6 | minor | U09-052 uses the same DE, x y' = y + sqrt(x^2+y^2) (written as x dy - y dx = sqrt(x^2+y^2) dx in 066), and its key is the separated form that 066 starts from. The families are not linked. | **aliased** Confirmed with sympy: U09-B-019 (x dy - y dx = sqrt(x^2+y^2) dx) reduces under y = vx to dv/sqrt(1+v^2) = dx/x, which is the key of U09-B-005. Related task (005 asks only for the separated form; 019 needs integration, the initial condition y(1) = 0 and y(3) = 4), not a duplicate. Alias requested [U09-B-005, U09-B-019]. U09-B-042 uses sqrt(x^2 - y^2), a different equation, so it is not included. | - |

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| ERR-162 | U09-B-041 JEE-MATH-U09-088 | E6 | minor | 088 is U09-064 with x and y swapped: dx/dy = x/y + y/x against dy/dx = x/y + y/x. U09-051 also uses this same DE, dy/dx = (x^2+y^2)/(xy), and its key is the first step of 064. None of the three items share a family dependency. | **aliased** Confirmed with sympy: U09-B-041 gives dx/dy = x/y + y/x, the x-y swap of U09-B-017 (dy/dx = x/y + y/x), and U09-B-004's key x v' = 1/v is 017's first step. Data and unknowns differ (017: y(1) = 2, find y(e); 041: y(1) = 1, find x0 with y(x0) = e, testing the choice of x = vy; 004: transformed equation only). Alias requested [U09-B-004, U09-B-017, U09-B-041]. | - |
| ERR-163 | U09-B-036 JEE-MATH-U09-083 | E6 | minor | The geometric condition in 083 reduces to exactly the DE of U09-082, 2xy y' = y^2 - x^2, with family x^2 + y^2 = Cx. 082 hands a student the family 083 needs, and the families are not linked. | **aliased** Confirmed with sympy: the tangent condition in U09-B-036 gives y' = (y^2 - x^2)/(2xy), the equation of U09-B-035, family x^2 + y^2 = Cx. Different points and unknowns (035: radius 5 through (1,3); 036: beta = 1 through (2,1)), and 036 adds the geometric translation step. Alias requested [U09-B-035, U09-B-036]. | - |
| ERR-164 | U09-B-030 JEE-MATH-U09-077 | E6 | minor | Statement I of 077 differs from U09-082 only by the coefficient 3, and its family 3x^2 + y^2 = Cx is 082's x^2 + y^2 = Cx rescaled, so the same task appears with trivially different numbers. The two do not share a family dependency. | **aliased** Confirmed: Statement I of U09-B-030 (family 3x^2 + y^2 = Cx) is a rescaling of U09-B-035 (x^2 + y^2 = Cx). Different task (judging a claimed general solution inside a two-statement item, vs the radius of a particular circle). Alias requested [U09-B-030, U09-B-035]. | - |
| ERR-165 | U09-B-026 JEE-MATH-U09-073 | E6 | minor | 073's equation 2xy y' = 2x^2 + y^2 is U09-054's 2xy y' = x^2 + y^2 with one coefficient changed. 073's distractor D even uses 054's family x^2 - y^2 = Cx. 054 is not linked to 073 (073 is linked only to 072). | **aliased** Confirmed with sympy: U09-B-026 (2xy y' = 2x^2 + y^2, family 2x^2 - y^2 = Cx) and U09-B-007 (2xy y' = x^2 + y^2, family x^2 - y^2 = Cx) differ in one coefficient. Different task (007: general solution; 026: curve through (3,6), find k at x = 1, k = 2 sqrt 2). Alias requested [U09-B-007, U09-B-026]. | - |
| ERR-166 | U09-B-045 JEE-MATH-U09-092 | E6 | minor | Statement II of 092, x^2 y' = xy - y^2, is U09-070's x^2 y' = xy + y^2 with one sign flipped, and it needs the same steps (x v' = -v^2 against v^2). The families are not linked. | **aliased** Confirmed with sympy: Statement II of U09-B-045 has solution y = x/(1 + log x) and U09-B-023 has y = x/(1 - log x); same steps up to one sign. Different tasks (023: value y(sqrt e) = 2 sqrt e; 045 II: blow-up as x -> 1/e from the right, inside a two-statement item). Alias requested [U09-B-023, U09-B-045]. | - |
| ERR-167 | U09-C-032 JEE-MATH-U09-126 | E6 | minor | Statement II of 126 asks for the integrating factor of y' + y tan x = sec x. That is exactly the standard form of U09-109's equation (cos x y' + y sin x = 1), and 109's explanation states the factor sec x. The families are not linked. | **aliased** Confirmed overlap is partial: Statement II of U09-C-032 is the integrating factor step of U09-C-015, whose asked value y(pi/3) is not given away. Related task, not a duplicate, so aliased. | - |
| ERR-168 | U09-C-019 JEE-MATH- | E6 | minor | After its geometric translation, 113 is y' - y/x = -x, which is U09-112's y' - y/x = x with the sign of the forcing flipped. Both use the factor 1/x, a point at x = 1 and evaluation at x = 3. The families are not linked. | **aliased** Confirmed U09-C-019 reduces to y' - y/x = -x, mirroring U09-C-018 (y' - y/x = x); different geometric translation and answer (-3 vs 6). Related task, aliased. | - |

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| | U09-113 | | | | | |
| ERR-169 | U09-C-003 JEE-MATH-U09-097 | E6 | minor | 097 (y' + y = e^{-x}) and U09-105 (y' - y = e^x) are mirror images of the same resonant-forcing case. 105's key form (x + C)e^{x} transfers straight to 097's (x + C)e^{-x}. The families are not linked. | **aliased**<br>Confirmed U09-C-003 (y' + y = e^{-x}, IVP) and U09-C-011 (y' - y = e^x, general solution) are mirror resonant cases; different tasks and answers. Aliased. | - |
| ERR-170 | U09-C-016 JEE-MATH-U09-110 | E6 | minor | 110's initial value problem (y' + y = 2x, y(0) = 1, so y(1) = 3/e) is the one U09-103's distractor B computes. 103 and 110 together show the full solution of that IVP. The families are not linked. | **aliased**<br>Confirmed U09-C-009's distractor value 3/e and rationale give the solution of U09-C-016's IVP (y' + y = 2x, y(0) = 1). Neither item's key is the other's answer; aliased so they never share a paper. | - |
| ERR-171 | U09-A-035 JEE-MATH-U09-035 | E6 | minor | Statement I of 035 checks the family of dy/dx = e^{x+y}. U09-021, dy/dx = e^{x+y} + x^2 e^y, has as its key the same family plus x^3/3, so it gives away the judgement. The families are not linked. | **aliased**<br>Confirmed relation: 021's key with the x^2 term removed gives the correct family for Statement I. It is a related task (different equation, one half of a statement item), not a duplicate, so no rewrite; alias group requested with U09-A-021. | - |
| ERR-172 | U09-A-017 JEE-MATH-U09-017 | E3 | minor | The explanation refers to options by position, and the reference is wrong: 'the last has degree 3', but the last option (D, the cube-root equation) has degree 1. The degree-3 equation is option C. Positional wording also breaks if options are shuffled. | **fixed**<br>Confirmed: 'the last has degree 3' was positional and wrong (the cube-root equation, degree 1, was last). Replaced with the equation itself: (y'')^3 = (y')^2 + y has degree 3. Stem, options, key unchanged. | - |
| ERR-173 | U09-A-025 JEE-MATH-U09-025 | E5 | minor | Distractor A (10 minutes) contradicts the stem, which says the temperature is 50 C at 10 minutes, not 35 C. A careful reader rules it out without solving. | **fixed**<br>Confirmed: distractor 10 contradicts the stem (the body is at 50 C at t = 10). Replaced with 30: counting excess temperatures 60, 30, 15 as three 10-minute stages instead of two halvings. Value updated; key 20 re-verified with sympy. | U09-A-025 |
| ERR-174 | U09-C-008 JEE-MATH-U09-102 | E5 | minor | Distractor B (20) is below the room temperature of 25 C, which a cooling body can never reach, so it is ruled out on sight. | **fixed**<br>Confirmed 20 is below the room temperature 25. Replaced it in U09-C-008 with 65: finds the excess 60e^{-t/10} = 20 correctly but treats it as the fall, 85 - 20 = 65 (checked in sympy). Values updated. | U09-C-008 |

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| ERR-175 | U09-C-033 JEE-MATH-U09-127 | E5 | minor | Distractor B (30) equals the room temperature, which the body only approaches as t goes to infinity and never reaches at t = 20, so it is ruled out on sight. | **fixed** Distractor 30 (room temperature) removed by the rewrite of U09-C-033. New distractors 7, 10, 46 each come from one named slip (double-counted GP sum, decay ratio taken on temperatures not excess, falls taken in AP) and all lie below 52, verified in sympy. | U09-C-033 |
| ERR-176 | U09-C-022 JEE-MATH-U09-116 | E5 | minor | The distractor rationales for A and B do not produce their options: they say 'miscomputes the particular solution' and '$1 - e^{-2}$ or similar' instead of naming one concrete slip. | **fixed** U09-C-022 rationales rewritten to single concrete slips: (I true, II false) takes y = Ce^{-x^2}, so y(0) = 0 forces y = 0 and y(1) = 0; (both false) solves 0 = 1 + C as C = 1, giving y(1) = 1 + 1/e. Both verified. | - |
| ERR-177 | U09-C-012 JEE-MATH-U09-106 | E5 | minor | The distractor B rationale offers two alternative slips ('$e^{x}$ or $2x$') instead of one specific error that leads to the option. | **fixed** U09-C-012 'both false' rationale now one slip: writes the factor as e^{P} = e^2 without integrating, rejecting e^{2x}. Also tightened the 'I false, II true' rationale to one slip (using e^{-int P dx}, which gives e^{-2x} and x). | - |
| ERR-178 | U09-C-008 JEE-MATH-U09-102 | E9 | minor | Temperature notation is inconsistent within the unit: 102 and 127 write 'degrees Celsius' in words, while 025 uses $^\circ\mathrm{C}$. | **fixed** U09-C-008 now writes $^\circ\mathrm{C}$; rewritten U09-C-033 uses $^\circ\mathrm{C}$ too, matching U09-A-025. | U09-C-008 |
| ERR-179 | U10-B-045 JEE-MATH-U10-095 | E6 | major | Same task as U10-044 with trivially different data: minimise PA + PB for P on a fixed line, with A and B on the same side, by reflecting one point in the line. 044 uses the x-axis as the mirror, 095 the line x + y = 2. Different concept families (U10-A-reflection-min-distance vs U10-B-min-path) and no shared family_dependencies entry, so both can be drawn into one paper. | **rewritten** Confirmed: 095 (min PM + PN on x + y = 2, same side, reflect one point) was U10-A-044's task with new data. Rewritten into a maximum problem with points on opposite sides: P on x + y = 2, M(1, 4), N(4, -4), maximum of \|PM - PN\|. Reflect N to N'(6, -2) on M's side; max = MN' = sqrt(61), attained at (16, -14) where line MN' meets the line (PM = 3 sqrt(61), PN = 2 sqrt(61)). Verified by sympy and a numeric scan. Distractors: sqrt(73) (bound MN, not attainable for opposite sides), sqrt(101) (wrong sign of lambda), sqrt(65) (foot in place of image). New family U10-B-max-diff-reflection; numerical alternate mu^2 = 61. Still Difficult. | U10-B-045 |
| ERR-180 | U10-B-033 JEE-MATH-U10-083 | E6 | major | Same task as U10-077: the distance from a point to a line measured along a given direction, found with the distance (symmetric) form (x1 + r cos t, y1 + r sin t). 077 gives the direction as inclination 45 degrees, 083 as 'parallel to 3x - 4y = 0'; both target lines are x + 2y = c. 083 (U10-B-distance-along-direction) is not linked to 077's family U10-B-distance-form. | **rewritten** Confirmed: 083 and 077 both computed a distance along a given direction with the distance form. Rewritten with the unknown reversed: a line through P(1, 1), not parallel to the x-axis, meets 2x + y = 13 at Q with PQ = 5; find the slope of PQ. 2cos t + sin t = 2 gives cos t = 1 (horizontal, excluded) or (3/5, 4/5), slope 4/3 (sympy solve over the unit circle, exactly one admissible root). Distractors 3/4 (cos/sin swapped), 1/2 (assumes PQ perpendicular), 4/5 (reports sin t). New family U10-B-direction-from- | U10-B-033 |

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| | | | | | distance; numerical alternate 3m = 4. Still Medium; question_type now Inverse/parameter MCQ. | |
| ERR-181 | U10-B-020 JEE-MATH-U10-070 | E6 | major | Same task as U10-061 with different numbers: given the foot Q of the perpendicular from a point P to a line L, write L as the line through Q perpendicular to PQ. 061 has P at the origin and asks for L; 070 has P(5, -1) and asks for L's x-intercept, one extra substitution. Families U10-B-foot-from-origin and U10-B-foot-inverse are not linked. | **rewritten** Confirmed: 070 repeated 061's task (line from a point and its foot) plus one substitution. Rewritten to ask for the image of the origin in L (the auditor's suggested twist): L is 2x - y - 1 = 0, image (4/5, -2/5), checked with sympy and the midpoint test. Distractors: (2/5, -1/5) (foot formula), (6/5, 12/5) (L along PQ), (-4/5, 2/5) (dropped minus sign). New family U10-B-foot-then-image. Still Medium; question_type now Multi-step computation MCQ. | U10-B-020 |
| ERR-182 | U10-C-010 JEE-MATH-U10-110 | E6 | major | Near-duplicates across two families. (a) U10-019 and U10-110 both ask for the radius of the circle through three given vertices, (0,0),(6,0),(2,4) vs (0,0),(4,0),(0,6). (b) U10-137 asks for the circle through (0,0),(4,0),(0,3), the vertices of the triangle cut off by the axes. That is 110's task with different intercepts, and both explanations use the same right-angle/diameter argument. 110 (U10-C-three-points) is linked to 117 but not to 019 (U10-A-circumradius) or 137 (U10-C-right-triangle-circumcircle). 019 is also the same circumcentre computation as 117 (U10-C-three-points). | **rewritten** Confirmed: radius of the circle through three given points including the origin repeats U10-A-019's task, and the right-angle/diameter route repeats U10-C-037. Rewrote U10-C-010: circle x^2 + y^2 + px + qy - 20 = 0 through (4, 2) and (-2, 2), find the radius (key 5; p = -2, q = 4, centre (1, -2)); no axis points, no right angle, unknown coefficients rather than three points. New family U10-C-coefficients-from-points; still Easy; integer numerical alternate added. Also requested alias U10-C-017 with U10-A-019 (same circumcentre computation). | U10-C-010 |
| ERR-183 | U10-A-036 JEE-MATH-U10-036 | E6 | major | 036 uses the same triangle as U10-137, (0,0),(4,0),(0,3). Its explanation says the circumcentre is (2, 3/2), the midpoint of the hypotenuse, which is the centre of 137's required circle x^2 + y^2 - 4x - 3y = 0. Its two statements also repeat U10-011 (circumcentre of a right triangle = midpoint of the hypotenuse) and U10-012 (orthocentre = right-angled vertex). None of these families is linked to U10-A-right-triangle-centres. | **rewritten** Confirmed: 036 used the triangle (0,0),(4,0),(0,3) of U10-C-037 and its statements repeated 011 (circumcentre of a right triangle) and 012 (orthocentre at the right-angled vertex). Rewritten into a different question on the same subtopic (centres of a triangle), same difficulty (Medium) and type (Statement-based): the acute isosceles triangle (-3,0),(3,0),(0,4). Statement I asks whether the incentre lies between the centroid and circumcentre on x = 0 (false: circumcentre 7/8 < centroid 4/3 < incentre 3/2); Statement II claims circumradius 25/4 (false: R = 25/8, the 25/4 slip is abc/(2 Delta)). Key stays 'both false'; check recomputes centroid, incentre, circumcentre and R with sympy. New family U10-A-isosceles-centres-order. No right triangle and no shared data with 011, 012 or U10-C-037, so no alias needed. | U10-A-036 |
| ERR-184 | U10-A-045 JEE-MATH-U10-045 | E6 | minor | The core of 045 is the same as U10-028. In 045 the three given points are the midpoints of the sides of the triangle formed by the fourth vertices, so the answer is 4 times the area of the given triangle, exactly as in 028. The distractor sets match (1x area, 2x area). Families are not linked. | **aliased** Confirmed related, not duplicate: 045 (fourth vertices of a parallelogram; PQR is the medial triangle, area 4 x 11/2 = 22) and 028 (triangle from its side midpoints, area 4 x 13/2 = 26) share the medial-triangle area relation with different data and answers (sympy shoelace re-verified 22 and 26). Alias group requested; no rewrite. | - |
| ERR-185 | U10-B-002 JEE-MATH-U10-052 | E6 | minor | 052 is the midpoint (1:1) case of U10-029. In both, a point divides the intercept segment between the axes in a given ratio and the intercepts are found from the section formula. 029's distractor D ('treats the point as the midpoint') is 052's method. Families U10-A-intercepts-from-section and U10-B-intercept-form are not linked. | **aliased** Confirmed related, not duplicate: 052 (midpoint case, intercepts 4 and 6) and U10-A-029 (ratio 1:2, intercepts 9/2 and -12) use the same section-formula method with different data and answers. Alias group requested; no rewrite. | - |

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| ERR-186 | U10-A-023 JEE-MATH-U10-023 | E6 | minor | U10-003 is the special case of 023's task with the line y = 0: the ratio in which a line divides a segment, read through the sign of lambda as internal or external. The answer sets ('2 : 3 internally' vs '2 : 9 internally') and distractor logic (sign of -L1/L2) are parallel. Families are not linked. | **aliased** Confirmed related: 003 (x-axis divides segment, lambda = 2/3) is the special case y = 0 of 023's line-divides-segment task (lambda = 2/9); same lambda : 1 method and sign reading, different data and answers (both re-verified with sympy solve). Alias group requested; no rewrite. | - |
| ERR-187 | U10-C-045 JEE-MATH-U10-145 | E6 | minor | The first step of 145 (intersect x - y = 1 with x^2 + y^2 = 25) is U10-105's whole task (intersect y = x + 1 with x^2 + y^2 = 25) with the line reflected in y = x. 105's distractor B, (4,3) and (-3,-4), described as 'solves with the line y = x - 1', gives exactly the points that 145's explanation finds. Families are not linked. | **aliased** Confirmed: the intersection step of 145 gives (4, 3), (-3, -4), which is 105's task with the reflected line and equals 105's distractor. Related task, not a duplicate (145's unknown is the circle through P, Q and O). Alias U10-C-045 with U10-C-005. | - |
| ERR-188 | U10-C-044 JEE-MATH-U10-144 | E6 | minor | 144 reduces to 143's task: a chord of a circle centred at the origin subtends a given angle at O, and the line's parameter follows from the distance d = r cos(theta/2). 144 adds only the step 'maximum area needs angle POQ = 90 degrees'. Families U10-C-max-area-chord and U10-C-chord-angle-origin are not linked, but 142 and 143 share the latter. | **aliased** Confirmed: after the max-area step (angle 90 degrees), 144 is the chord-angle-at-origin computation of 143 (and related to 142). Alias U10-C-044 with U10-C-043 and U10-C-042. | - |
| ERR-189 | U10-C-040 JEE-MATH-U10-140 | E6 | minor | 140 generalises U10-139 with a parameter. Both count the points of an origin-centred circle at a fixed distance from a line, by comparing the two parallel lines at that distance with the radius (tangent / secant / miss casework). Families are not linked. | **aliased** Confirmed: same two-parallel-lines-versus-radius casework as 139, parametrised; key (-25,-15) U (15,25) re-verified. Alias U10-C-040 with U10-C-039. | - |
| ERR-190 | U10-A-034 JEE-MATH-U10-034 | E6 | minor | 034 is U10-018's task (orthocentre from vertices, by two altitudes) with a first step that finds the vertices from three lines. The distractor sets match (centroid, circumcentre, altitude drawn with the side's own slope). Families are not linked. | **aliased** Confirmed related: 034 (vertices from three lines, then orthocentre (2,2)) extends 018 (orthocentre from vertices, (1,1)) by one step; same two-altitude method and parallel distractor logic, different data and answers (both re-verified). Alias group requested; no rewrite. | - |
| ERR-191 | U10-C-033 JEE-MATH-U10-133 | E6 | minor | Both statements of 133 repeat other items. Statement I (position of (6,-2) by the sign of S1) is U10-104's task. Statement II (y = x + 4 meets x^2 + y^2 = 8 at exactly one point) is the tangency test of U10-106 and U10-111. 133's family U10-C-stmt-position-tangency is linked to neither U10-C-position-point nor U10-C-tangency-condition. | **aliased** Confirmed: Statement I is the S1 position test of 104 and Statement II the tangency test of 106/111 (distance 4/sqrt(2) = 2sqrt(2) = r). Aliases U10-C-033 with U10-C-004, U10-C-006 and U10-C-011 (as pairs, so 004 and 006/011 stay independent). | - |

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| ERR-192 | U10-C-029 JEE-MATH-U10-129 | E6 | minor | Statement I of 129 (length of the chord of x^2 + y^2 = 25 along x + y = 5) is the chord-length task of U10-109 and U10-120. Statement II (whether the circle on a given diameter passes through a point) uses the diameter form of 103 and 131. 129's family U10-C-stmt-chord-diameter is linked to neither U10-C-chord-length nor U10-C-diameter-form. | **aliased** Confirmed: Statement I (chord 5sqrt(2)) is the chord-length task of 109/120; Statement II uses the diameter form of 103. Aliases U10-C-029 with U10-C-009, U10-C-020 and U10-C-003 (pairs). | - |
| ERR-193 | U10-B-015 JEE-MATH-U10-065 | E6 | minor | Statement I of 065 (the distance between 3x - 4y + 2 = 0 and 6x - 8y + 9 = 0, with the rescaling trap) is U10-054's task with different numbers. U10-B-stmt-parallel is not linked to U10-B-dist-parallel (054, 080). | **aliased** Confirmed related: Statement I of 065 (distance 1/2 after rescaling) uses 054's rescaling method on different lines; Statement II (sign test) is a separate idea. Alias group with U10-B-004 and U10-B-030 (the dist-parallel family) requested; no rewrite. | - |
| ERR-194 | U10-B-029 JEE-MATH-U10-079 | E6 | minor | The two statements of 079 are the foot formula (U10-057) and the image formula (U10-058) on new numbers, with the same sign and factor traps. U10-B-stmt-foot-image is linked to neither U10-B-foot nor U10-B-image. | **aliased** Confirmed related: the statements of 079 apply the foot and image formulas (foot (8/5, 9/5), image of O (72/25, 96/25), both re-verified) on new data, as do 057 and 058. Two alias groups requested (079 with 057, 079 with 058) so that 057 and 058 can still appear together; no rewrite. | - |
| ERR-195 | U10-C-038 JEE-MATH-U10-138 | E6 | minor | 138's explanation computes the least distance between the line and the circle ('sqrt(5), which equals 3*sqrt(5) - 2*sqrt(5)'), which is U10-127's task. Both rest on the perpendicular from the centre to the line. Families are not linked. | **aliased** Confirmed: the closing distance check in 138's explanation computed 127's least-distance quantity. Removed that sentence from U10-C-038's explanation (it still reaches (4, 2)) and aliased U10-C-038 with U10-C-027, since both rest on the perpendicular from the centre to the line. | - |
| ERR-196 | U10-C-001 JEE-MATH-U10-101 | E5 | minor | The rationale for distractor D does not describe the slip that produces it. It says the student uses the coefficients -4 and 6 'as 2g, 2f with sign changed'. That gives g = -2, f = 3, which is the correct centre (2, -3) and r = 3. Option D, centre (4,-6) and r = 4*sqrt(3), comes from taking -4 and 6 as g and f themselves. | **fixed** Confirmed with sympy: g = -4, f = 6, c = 4 gives centre (4, -6), r^2 = 48 = (4sqrt(3))^2; the old wording (2g, 2f) gives the key. Rationale reworded: takes -4 and 6 themselves as g and f, forgetting the factor 2. | - |
| ERR-197 | U10-D-013 JEE-MATH-U10-163 | E6 | major | Item 153's stem is exactly item 163's key, so either item gives the other away; no shared family_dependencies. | **rewritten** Confirmed: 163's key 3y^2 = 16x was the stem of 153. U10-D-013 rewritten into a different task: vertex at origin, axis along the y-axis, focus located as the intercept of 3x + 2y - 12 = 0 on the axis; key x^2 = 24y. New distractors, explanation, sympy check and new family U10-D-eqn-focus-on-line. No longer shares any data or answer with 153 or 160. | U10-D-013 |

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| ERR-198 | U10-D-010 JEE-MATH-U10-160 | E6 | minor | Near-duplicate of 163: both find $4a$ by substituting one point into $y^2 = 4ax$ (only the reported form differs). No shared family. | **retained** Overlap was only with 163 (single substitution of a point into y^2 = 4ax). 163 (U10-D-013) has been rewritten into a focus-on-a-line task with the axis along the y-axis, so U10-D-010 no longer has a near-duplicate; no alias needed. | - |
| ERR-199 | U10-E-012 JEE-MATH-U10-212 | E6 | major | Same ellipse as 210 ($5x^2 + 9y^2 = 45$ is $\frac{x^2}{9} + \frac{y^2}{5} = 1$); 210's key $(2, \frac{5}{3})$ shows the focus at $x = 2$, which gives 212's key $2c = 4$ at once. No shared family. | **rewritten** Confirmed: 5x^2 + 9y^2 = 45 is the ellipse of U10-E-010, whose key (2, 5/3) shows c = 2 and so 2c = 4. Rewritten: the ellipse is now given by its axis lengths (major 34, minor 16), a = 17, b = 8, c = 15, key 30 (sympy via _ell). Distractors 15 (c only), 2sqrt(353) (c^2 = a^2 + b^2), 60 (full lengths in c^2 = a^2 - b^2, then doubled). New family U10-E-focal-gap-from-axes; numerical alternate 30. | U10-E-012 |
| ERR-200 | U10-E-032 JEE-MATH-U10-232 | E6 | major | 232's stem states 217's answer: it gives the ellipse $\frac{x^2}{16} + \frac{y^2}{12} = 1$ together with the focus $S(2, 0)$ and the line $x = 8$, which are exactly 217's data. No shared family. | **rewritten** Confirmed: the stem printed x^2/16 + y^2/12 = 1 with focus (2, 0) and line x = 8, which is exactly U10-E-017's data and key. Rebuilt on x^2/36 + y^2/32 = 1 (a = 6, c = 2, e = 1/3) and the unknown is reversed: P is 15 from the directrix x = 18, find PS' to the far focus (-2, 0). PS = 5 by the focus-directrix property, PS' = 12 - 5 = 7 (sympy solves for x = 3 and computes the distance). Distractors 5 (near focus), 9 (e dropped in a + ex), 31/3 (e^2 used for e). New family U10-E-directrix-to-far-focus. | U10-E-032 |
| ERR-201 | U10-E-037 JEE-MATH-U10-237 | E6 | major | Same ellipse as 227: foci $(0, \pm 4)$ with focal sum $10$; 237's key $25x^2 + 9y^2 = 225$ gives $b = 3$, i.e. 227's key (minor axis $6$), and 227's working gives 237. No shared family. | **rewritten** Confirmed: foci (0, +-4) with focal sum 10 is U10-E-027's ellipse (b = 3). The suggested foci (0, +-3) would reproduce U10-E-023's ellipse, so that suggestion was not used. Rewritten with a shifted centre: foci (1, 2) and (1, -4), sum 10, key (x - 1)^2/16 + (y + 1)^2/25 = 1 (centre (1, -1), a = 5, c = 3, b = 4). The check removes the radicals and verifies sample points give a sum of exactly 10. Distractors: wrong orientation, a sign slip in the shift, and c^2 used for b^2. New family U10-E-locus-focal-sum-shifted. Translated conics are in scope per the E8 ruling. | U10-E-037 |
| ERR-202 | U10-E-001 JEE-MATH-U10-201 | E6 | major | 201 asks the eccentricity of $\frac{x^2}{25} + \frac{y^2}{16} = 1$, but 213, 221 and 230 all print that same ellipse with its focus $(3, 0)$ in the stem, which gives $e = \frac{3}{5}$ in one step. No shared family. | **rewritten** Confirmed: 013, 021 and 030 print x^2/25 + y^2/16 = 1 with focus (3, 0), which gives e = 3/5 at once. Rewritten as an eccentricity item with a vertical major axis: x^2/40 + y^2/49 = 1, e = 3/7. Distractors 2sqrt(10)/7 (b/a), 3sqrt(10)/20 (c/b), 9/49 (e^2). New family U10-E-ecc-vertical. Still Easy. | U10-E-001 |
| ERR-203 | U10-F-002 JEE-MATH-U10-252 | E6 | major | 252 uses the same hyperbola as 251 ($9x^2 - 16y^2 = 144$ is $\frac{x^2}{16} - \frac{y^2}{9} = 1$); 252's key $(\pm 5, 0)$ with $a = 4$ gives 251's key $e = \frac{5}{4}$, and vice versa. No shared family. | **rewritten** Confirmed: 9x^2 - 16y^2 = 144 is 251's curve and the foci (+-5, 0) with a = 4 give 251's key. Rewritten to a different curve and unknown: distance between the foci of 5x^2 - 4y^2 = 20 (x^2/4 - y^2/5 = 1, c = 3, key 6; sympy via _hyp foci). New distractors 2 (ellipse relation), 4 (2a), 3 (c); numerical alternate 6 added. New family U10-F-focal-separation. | U10-F-002 |

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| ERR-204 | U10-F-012 JEE-MATH-U10-262 | E6 | major | 262 is the exact inverse of 251 with identical numbers: 251 gives $\frac{x^2}{16} - \frac{y^2}{9} = 1$ and asks $e$ ($\frac{5}{4}$); 262 gives $\frac{x^2}{16} - \frac{y^2}{b^2} = 1$ with $e = \frac{5}{4}$ and asks $b^2$ ($9$). Each item states the other's key. No shared family. | **rewritten** Confirmed: exact inverse of 251 with the same numbers. Rewritten with a different unknown and new data: x^2/a^2 - y^2/45 = 1 with e = 3/2, find a^2 = 36 (sympy solve). Distractors 90 (e not squared), 225/4 (inverted relation), 6 (a for a^2); numerical answer 36. New family U10-F-transverse-from-ecc. Curve x^2/36 - y^2/45 is not used elsewhere in U10. | U10-F-012 |
| ERR-205 | U10-F-033 JEE-MATH-U10-283 | E6 | major | Near-duplicate of 260: 283's hyperbola has vertices $(\pm 3, 0)$ and foci $(\pm 5, 0)$, exactly 260's data, so 260's key ($b^2 = 16$) leaves 283 one formula away. The same curve $\frac{x^2}{9} - \frac{y^2}{16} = 1$ also appears in the stems of 270, 292 and 293. No shared family. | **rewritten** Confirmed: the 25/16 ellipse gave a = 3, c = 5, exactly 260's data. Rewritten with ellipse x^2/289 + y^2/225 = 1 (hyperbola a = 8, c = 17) and a different unknown, the distance between the hyperbola's directrices, 2a/e = 128/17 (sympy via _hyp directrices). Distractors 64/17 (a/e), 289/4 (ellipse directrices), 34 (2ae). New family U10-F-from-ellipse-directrix. The suggested 169/144 ellipse was avoided because it reproduces 254's curve x^2/25 - y^2/144. | U10-F-033 |
| ERR-206 | U10-F-020 JEE-MATH-U10-270 | E6 | major | 292's stem gives away 270: it states the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$, its directrix $x = \frac{9}{5}$ and its focus $(5, 0)$, so 270's key is $5 - \frac{9}{5} = \frac{16}{5}$. 270's stem also prints 260's key equation. No shared families. | **rewritten** Confirmed: 292's stem states the directrix 9/5 and focus (5, 0), giving 270's key. Rewritten as a reversed problem with no fixed curve: e = 3 and focus-to-nearer-directrix distance 8, find the latus rectum. 8a/3 = 8 gives a = 3, b^2 = 72, LR = 48 (sympy solve plus _hyp lr and a gap assertion). Distractors 192/5 (farther directrix), 24 (semi-latus rectum), 64 (focus-to-vertex). Numerical alternate 48 added. New family U10-F-focus-directrix-lr. The suggested 36/64 curve was avoided because it is 258's curve. | U10-F-020 |
| ERR-207 | U10-E-050 JEE-MATH-U10-250 | E6 | major | 211's stem nearly gives away 250's key: both use the same form '$\frac{x^2}{16} + \frac{y^2}{b^2} = 1$, $0 < b < 4$', and 211 states that this ellipse has eccentricity $\frac{\sqrt{3}}{2}$ (b = 2), which is 250's key. No shared family. | **rewritten** Confirmed: 011 states that x^2/16 + y^2/b^2 has e = sqrt(3)/2, which is 050's key. Rewritten with E1 = x^2/169 + y^2/b^2 and e2 = 12/13, and a new unknown, the latus rectum of E2. The key step is unchanged: the semi-major axis of E2 is sqrt(c1^2 + b^2) = 13. That gives b = 12, c1 = 5, so E2 is x^2/25 + y^2/169 = 1 with latus rectum 50/13. Distractors 288/13 (E1's latus rectum), 25/13 (semi-latus rectum), 338/5 (axes swapped). Numerical alternate 13l = 50. New family U10-E-two-ellipses-lr. | U10-E-050 |
| ERR-208 | U10-F-035 JEE-MATH-U10-285 | E6 | minor | 264's stem $16y^2 - 9x^2 = 576$ is 285's key $\frac{y^2}{36} - \frac{x^2}{64} = 1$ in unreduced form (264's own working finds the foci $(0, \pm 10)$, 285's data). No shared family. | **fixed** Confirmed: 264's stem 16y^2 - 9x^2 = 576 is 285's key unreduced. Changed 285's data to foci (0, +-15), e = 5/3: a = 9, b^2 = 144, key y^2/81 - x^2/144 = 1; distractors rebuilt with the same slips (orientation, swap, b^2 = a^2 + c^2 = 306). The suggested foci (0, +-13), e = 13/5 was avoided because it gives the conjugate in 256 and 256's key. | U10-F-035 |
| ERR-209 | U10-F-048 JEE-MATH-U10-298 | E6 | minor | Statement I of 298 is the same task as 256 (eccentricity of the conjugate of a given hyperbola) with different numbers; the families are not linked. | **aliased** Confirmed related, not a duplicate: Statement I of 298 (conjugate of x^2/12 - y^2/4, e' = 2) uses the same method as 256 (conjugate of x^2/144 - y^2/25, e' = 13/5) with different data, and 298 also tests the foci-square fact. Alias group requested; no rewrite. | - |

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| ERR-210 | U10-E-029 JEE-MATH-U10-229 | E6 | minor | 229 contains 209 in full: 209's whole condition (distance between the foci equals the minor axis, so $b = c$, $e = \frac{1}{\sqrt{2}}$) is the first step of 229. No shared family. | **aliased** Confirmed related, not duplicate: 229 uses 209's shape condition (b = c) as its first step, but then asks for the major axis from the latus rectum 10 (key 20). That is a different unknown and needs more work. Alias requested for U10-E-009 and U10-E-029; no rewrite. | - |
| ERR-211 | U10-F-016 JEE-MATH-U10-266 | E6 | minor | 266's key $\frac{x^2}{16} - \frac{y^2}{9} = 1$ (foci $(\pm 5, 0)$) is printed as the stem curve of 272 (and is 251's curve, 252 unreduced), so a student can match option D on sight. | **fixed** Confirmed: 266's key x^2/16 - y^2/9 = 1 is printed in 272's stem. Changed 266 to foci (+-6, 0) through P(6, 11/5): PS = 11/5, PS' = 61/5, a = 5, b^2 = 11, key x^2/25 - y^2/11 = 1 (sympy). Distractors rebuilt (swap, b = c, b^2 = a^2 + c^2). The 16/9 curve now remains only in 251 and 272, neither of which states the other's key. | U10-F-016 |
| ERR-212 | U10-E-007 JEE-MATH-U10-207 | E6 | minor | 207's key $\frac{x^2}{25} + \frac{y^2}{9} = 1$ (vertices $(\pm 5, 0)$, foci $(\pm 4, 0)$) is printed in the stems of 215, 225 and 240, so option D is recognisable on sight. | **fixed** Confirmed: the key x^2/25 + y^2/9 = 1 is printed in the stems of 015, 025 and 040. The data is now vertices (+-10, 0) and foci (+-8, 0), key x^2/100 + y^2/36 = 1. The suggested 13/12/5 data was avoided because 050 now uses a = 13. Distractors are re-derived: c^2 for b^2 gives 64, a^2 + c^2 gives 164, and the axes swapped gives x^2/36 + y^2/100. No other stem in the batch prints this ellipse. | U10-E-007 |
| ERR-213 | U10-F-047 JEE-MATH-U10-297 | E5 | minor | Two distractors can be ruled out by symmetry: no lattice point of $x^2 - y^2 = 45$ lies on an axis, so the points come in groups of four $(\pm x, \pm y)$ and the count must be a multiple of 4. That rules out $6$ (B) and $3$ (C) without any work. | **fixed** Confirmed: no lattice point lies on an axis, so the count is a multiple of 4 and 6 and 3 were ruled out on sight. Replaced them with 8 (discarding the factorisation 1 x 45, losing (+-23, +-22)) and 16 (divisors 1, 3, 5, 9 with four signs each, counting (+-7, +-2) twice). Key 12 unchanged (brute-force enumeration). | U10-F-047 |
| ERR-214 | U10-E-048 JEE-MATH-U10-248 | E5 | minor | Option D ($4$) can be ruled out on sight: the admissible $c$ form a symmetric interval $(-r, r)$ containing $0$, so the number of integers must be odd. | **fixed** Confirmed: the admissible c form a symmetric interval about 0, so the count must be odd and 4 was ruled out on sight. Replaced it with 1, which comes from dropping the cross term: (2x + c)^2 = 4x^2 + c^2 gives 17x^2 + 4c^2 - 4 = 0, which needs c^2 < 1, so only c = 0. Key 9 is unchanged (brute-force check). The misconception text is updated to match. | U10-E-048 |
| ERR-215 | U10-F-025 JEE-MATH-U10-275 | E8 | minor | Shifted (non-central) hyperbola. The U10 scope brief says conics 'in standard forms', and the U10-F authoring brief limits hyperbolas to 'standard position'; shifted forms are licensed only for the parabola (batch D). | **retained** Errata ruling: translated (shifted) standard conics are in scope. Completing squares only; no out-of-unit tools in the explanation. | - |
| ERR-216 | U10-F-026 JEE-MATH- | E8 | minor | Shifted (non-central) hyperbola. The U10 scope brief says conics 'in standard forms', and the U10-F authoring brief limits hyperbolas to 'standard position'; shifted forms are licensed only for the parabola (batch D). | **retained** Errata ruling: translated (shifted) standard conics are in scope. Completing squares only; no out-of-unit tools in the explanation. | - |

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| | U10-276 | | | | | |
| ERR-217 | U10-F-034 JEE-MATH-U10-284 | E8 | minor | Shifted (non-central) hyperbola. The U10 scope brief says conics 'in standard forms', and the U10-F authoring brief limits hyperbolas to 'standard position'; shifted forms are licensed only for the parabola (batch D). | **retained** Errata ruling: translated (shifted) standard conics are in scope. Completing squares only; no out-of-unit tools in the explanation. | - |
| ERR-218 | U10-F-046 JEE-MATH-U10-296 | E8 | minor | Shifted (non-central) hyperbola. The U10 scope brief says conics 'in standard forms', and the U10-F authoring brief limits hyperbolas to 'standard position'; shifted forms are licensed only for the parabola (batch D). It also needs the centre found by intersecting two oblique asymptotes. | **retained** Errata ruling: translated (shifted) standard conics are in scope. Finding the centre by intersecting the two asymptotes is intersection of lines, which is in Unit 10, so the explanation uses no out-of-unit tools. | - |
| ERR-219 | U10-F-050 JEE-MATH-U10-300 | E8 | minor | Shifted (non-central) hyperbola. The U10 scope brief says conics 'in standard forms', and the U10-F authoring brief limits hyperbolas to 'standard position'; shifted forms are licensed only for the parabola (batch D). | **retained** Errata ruling: translated (shifted) standard conics are in scope. Uses the focal-difference definition, midpoint and c^2 = a^2 + b^2 only. | - |
| ERR-220 | U11-C-004 JEE-MATH-U11-098 | E6 | major | Same task as U11-065: two Cartesian lines with an unknown in the point of the second line ('y-k' / 'y+k') are said to intersect; find k. The only differences are the numbers and the method shown (determinant vs solving for parameters). The families are different and have no shared family_dependencies entry, so the generator can put both in one paper. | **rewritten** Confirmed with sympy: old U11-C-004 (k = 8) was the same task as U11-B-018 (k = -2), an unknown in the point of a second line made to intersect the first, with only the numbers changed. Rewritten into a different task on the same subtopic and difficulty (Easy, Inverse/parameter): find k so that the line (x-1)/2 = (y-k)/3 = (z+2)/1 meets the x-axis (key k = -6, common point (5,0,0)), solved by imposing y = z = 0 on the general point; check also confirms via the coplanarity test against the x-axis. New family U11-C-meets-axis-param. Distractors 6, 3/2, -2/3 each reproduced from the named slip. Numerical alternate dropped (negative key); batch still has 12. | U11-C-004 |
| ERR-221 | U11-C-016 JEE-MATH-U11-110 | E6 | major | Same task as U11-075: two Cartesian lines with an unknown direction ratio in the first line are said to intersect; find the unknown. U11-075 and U11-065 share 'U11-B-intersect-param', but 110 is in a separate family with no link to either, so 075 and 110 (or 065, 098 and 110) can appear in the same paper. | **rewritten** Confirmed with sympy: old U11-C-016 (lambda = 3) was the same task as U11-B-028 (k = 2), an unknown direction ratio fixed by the intersection condition. Batch C already has quadratic-parameter intersection items (C-025, C-040), so the rewrite uses a different unknown and twist: lambda is fixed by perpendicularity (b1.b2 = 0 gives lambda = 2), then the skew lines' shortest distance is asked (key 2; triple product 18, \|b1 x b2\| = 9). Still Medium. New family U11-C-perp-param-sd, topic moved to shortest distance, type Multi-step computation. Distractors 6 sqrt(65)/65, sqrt(6), 2/9 verified. Integer numerical alternate (answer 2) added. | U11-C-016 |
| ERR-222 | U11-B-025 | E6 | major | Same task as U11-044: the acute angle between two lines, where one direction ratio is a parameter k, is given (pi/3 in 044, pi/4 in 072). Squaring | **rewritten** Confirmed with sympy: old U11-B-025 (DRs (2,-1,2),(-1,k,0), acute angle pi/4, 7k^2- | U11-B-025 |

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| | JEE-MATH-U11-072 | | | the cosine condition gives a quadratic and both roots must be kept. 044 asks for the set and 072 for the sum, but the method and the trap (do not reject a root) are the same. Different families with no shared dependency. | 8k+1=0, sum 8/7) repeats U11-A-044's method and keep-both-roots trap. Rewrote U11-B-025: perpendicularity to L1 fixes a = 2 in L2's direction (-1,a,2), then cos of the acute angle between L2 and a third line L3 with direction (2,3,6) is 16/21. New unknown and twist, no quadratic; distractors 4/21 (sign slip a = -2), 13/21 (angle of L1 with L3), 20/21 (dropped sign of -i) each verified. Parameter renamed to a to avoid clash with k-hat. New family U11-B-perp-then-angle; still Medium, XII; numerical alternate 21 cos(theta) = 16. | |
| ERR-223 | U11-B-033 JEE-MATH-U11-080 | E6 | minor | Close to U11-062. Both items find the foot of the perpendicular from a point to a Cartesian line and report alpha+beta+gamma. 080 only adds the step 'nearest point = foot'. Different families with no shared dependency. | **aliased**<br>Verified (sums 2 and 10). Related task (nearest point = foot) with different data and answer, not a duplicate; alias U11-B-033 with U11-B-015. | - |
| ERR-224 | U11-B-021 JEE-MATH-U11-068 | E6 | minor | Close to U11-058. 068 asks for PQ, where Q is the image of P in a line. This is twice the perpendicular distance that 058 asks for, found the same way (foot, then distance). Different families with no shared dependency. | **aliased**<br>Verified (3 and 6 = 2 x 3). Same foot-then-distance method with different data; alias U11-B-021 with U11-B-011. | - |
| ERR-225 | U11-B-032 JEE-MATH-U11-079 | E6 | minor | Close to U11-009: three points with unknown coordinates are said to be collinear, and the unknown(s) are found from proportional direction ratios. Different families with no shared dependency. | **aliased**<br>Verified (k = 5; k + m = 16). Same collinearity method with different data and unknowns; alias U11-B-032 with U11-A-009. | - |
| ERR-226 | U11-B-014 JEE-MATH-U11-061 | E6 | minor | Close to U11-018: a perpendicularity condition with an unknown k in the first direction ratio. 061 even uses the same first direction triple (k, 2, -3) as 018. 018 leads to a quadratic and 061 is linear, but the task and the condition are the same. Different families with no shared dependency. | **aliased**<br>Verified (U11-A-018 sum -1 via quadratic; U11-B-014 k = 2, linear). Same perpendicularity condition, different second direction and answer; alias U11-B-014 with U11-A-018. | - |
| ERR-227 | U11-C-009 JEE-MATH-U11-103 | E6 | minor | Close to U11-002. Once you notice the line is parallel to the z-axis, 103 becomes 'distance of the point (3, -4, z) from the z-axis', which is sqrt(x^2+y^2). This is the same one-step task as 002 (distance of a point from an axis). Families differ: 103 is in 'U11-C-sd-axes', 002 in 'U11-A-distance-axis'. | **aliased**<br>Confirmed related, not duplicate: U11-C-009 (distance between the z-axis and the line x = 3, y = -4, key 5) reduces to the distance of a point from an axis, as in U11-A-002 (distance of (3,-4,12) from the y-axis, sqrt(153)). Different data, axis and answer, and C-009 adds recognising parallel lines, so no rewrite. Alias group requested. | - |
| ERR-228 | U11-A-028 JEE- | E5 | minor | The rationale for distractor B does not produce option B. Applying the stated misconception (testing l+m+n=1) rejects both statements, which is option D, | **fixed**<br>Verified: testing l+m+n=1 rejects both statements (2.073 and sqrt 2), giving the both-false option. Rationales rewritten: that slip now drives the both-false distractor; the l- | - |

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| | MATH-U11-028 | | | not B. The rationale has to add 'accepts Statement I without testing it' to reach B, so it is an ad hoc patch rather than one consistent error. | true-II-false distractor now comes from one consistent wrong criterion (requiring every direction angle strictly between 0 and 90 degrees without testing the identity). | |
| ERR-229 | U11-C-011 JEE-MATH-U11-105 | E5 | minor | Option D ('coincident') can be ruled out on sight, because the direction ratios (2,1,-1) and (1,3,2) are visibly not proportional. The distractor review itself admits it 'attracts only a student who guesses'. So the item is effectively a 3-option MCQ. | **fixed**<br>Confirmed: the coincident option was ruled out on sight and its rationale admitted it caught only guessers. Second line changed to (x-2)/1 = (y+1)/(-1) = z/1 so the directions (2,1,-1), (1,-1,1) are perpendicular (dot product 0) while the triple product is 9 (skew). Options are now 'skew and perpendicular' (key), 'skew but not perpendicular' (belief that perpendicular lines must meet), 'intersecting and perpendicular' (solves x and y equations, s = -1, t = -3, without checking z), 'parallel and distinct' (no common point taken as parallel). Mutually exclusive; check returns (nature, perpendicular) from sympy; explanation updated. | U11-C-011 |
| ERR-230 | U11-A-001 JEE-MATH-U11-001 | E7 | minor | source_class is 'XII', but the content (distance between two points) uses only coordinates in space and the 3D distance formula. NCERT places this in Class XI ('Introduction to Three Dimensional Geometry'), not in the Class XII 'Three Dimensional Geometry' chapter, which starts at direction cosines. | **fixed**<br>source_class set to XI (3D distance formula is NCERT XI). | - |
| ERR-231 | U11-A-002 JEE-MATH-U11-002 | E7 | minor | source_class is 'XII', but the content (distance of a point from an axis) uses only coordinates in space and the 3D distance formula. NCERT places this in Class XI ('Introduction to Three Dimensional Geometry'), not in the Class XII 'Three Dimensional Geometry' chapter, which starts at direction cosines. | **fixed**<br>source_class set to XI. | - |
| ERR-232 | U11-A-012 JEE-MATH-U11-012 | E7 | minor | source_class is 'XII', but the content (unknown coordinate from the distance formula) uses only coordinates in space and the 3D distance formula. NCERT places this in Class XI ('Introduction to Three Dimensional Geometry'), not in the Class XII 'Three Dimensional Geometry' chapter, which starts at direction cosines. | **fixed**<br>source_class set to XI. | - |
| ERR-233 | U11-A-020 JEE-MATH-U11-020 | E7 | minor | source_class is 'XII', but the content (triangle type from three distances) uses only coordinates in space and the 3D distance formula. NCERT places this in Class XI ('Introduction to Three Dimensional Geometry'), not in the Class XII 'Three Dimensional Geometry' chapter, which starts at direction cosines. | **fixed**<br>source_class set to XI. | - |
| ERR-234 | U11-A-021 JEE-MATH- | E7 | minor | source_class is 'XII', but the content (area via three distances) uses only coordinates in space and the 3D distance formula. NCERT places this in Class XI ('Introduction to Three Dimensional Geometry'), not in the Class XII 'Three Dimensional Geometry' chapter, which starts at direction cosines. | **fixed**<br>source_class set to XI. | - |

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| | U11-021 | | | | | |
| ERR-235 | U11-A-024 JEE-MATH-U11-024 | E7 | minor | source_class is 'XII', but the content (point on an axis equidistant from two points) uses only coordinates in space and the 3D distance formula. NCERT places this in Class XI ('Introduction to Three Dimensional Geometry'), not in the Class XII 'Three Dimensional Geometry' chapter, which starts at direction cosines. | **fixed**<br>source_class set to XI. | - |
| ERR-236 | U11-A-033 JEE-MATH-U11-033 | E7 | minor | source_class is 'XII', but the content (median length (midpoint and distance)) uses only coordinates in space and the 3D distance formula. NCERT places this in Class XI ('Introduction to Three Dimensional Geometry'), not in the Class XII 'Three Dimensional Geometry' chapter, which starts at direction cosines. | **fixed**<br>source_class set to XI. | - |
| ERR-237 | U11-A-045 JEE-MATH-U11-045 | E7 | minor | source_class is 'XII', but the content (quadrilateral type from distances) uses only coordinates in space and the 3D distance formula. NCERT places this in Class XI ('Introduction to Three Dimensional Geometry'), not in the Class XII 'Three Dimensional Geometry' chapter, which starts at direction cosines. | **fixed**<br>source_class set to XI. | - |
| ERR-238 | U11-A-003 JEE-MATH-U11-003 | E7 | minor | source_class 'XII' for the section-formula, midpoint and centroid items (003, 004, 009, 010, 013, 014, 025, 026, 027, 029, 038, 041, 043, 046) needs a policy decision. Coordinate section formula was in NCERT XI 3D (removed in the 2023 rationalisation). The only current NCERT home is the vector section formula in XII Vector Algebra, not XII 3D Geometry. All 22 'Coordinates of a point in space' items are tagged XII without a stated rule. | **fixed**<br>Rule applied per errata policy (3D coordinates, distance, section formula are Class XI): all 22 'Coordinates of a point in space' items in batch A (001-004, 009, 010, 012-014, 020, 021, 024-027, 029, 033, 038, 041, 043, 045, 046) now source_class XI. Direction-cosine and angle-between-lines items stay XII. | - |
| ERR-239 | U11-A-043 JEE-MATH-U11-043 | E9 | minor | The common_misconception has maths written as plain text ('1/3') outside $...$, unlike the rest of the slice. | **fixed**<br>Misconception rewritten with $\frac{1}{3}$ in LaTeX; key 8 re-verified (k = -6, -4/3). | - |
| ERR-240 | U12-C-016 JEE-MATH-U12-123 | E6 | major | Near-duplicate of U12-059 with no shared family_dependencies: both start from unit vectors with \|a^+b^\| = sqrt(3); U12-059 asks for the angle (pi/3), and U12-123 only adds \|a^ x b^\| = sin(theta). If both appear in one paper, U12-059's answer gives away U12-123. | **rewritten**<br>Confirmed: same hypothesis \|a^+b^\| = sqrt(3) as U12-B-005, whose answer pi/3 gives sin(pi/3) at once. Rewrote U12-C-016 as a new Easy item: \|a\| = 4, \|b\| = 5, scalar projection of b on a is 3, find \|a x b\| (= 16 via a.b = 12 and Lagrange's identity). New family U12-C-cross-from-projection; numerical alternate 16 added. | U12-C-016 |

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| ERR-241 | U12-B-039 JEE-MATH-U12-093 | E6 | minor | Shares the hypothesis \|a^+b^\| = sqrt(3) with U12-059 and U12-123 without a shared dependency. Its first step (a^.b^ = 1/2) is U12-059's whole task. | **aliased** Verified: \|a^+b^\| = sqrt 3 gives a^.b^ = 1/2 (sympy), which is the whole of U12-B-005's task and the first step of U12-C-016. U12-B-039 asks a different unknown (a four-term expansion, key -11/2), so it is a related task, not a duplicate; linked as an alias group with U12-B-005 and U12-C-016. | - |
| ERR-242 | U12-C-027 JEE-MATH-U12-134 | E6 | major | Same task as U12-152 Statement II with no shared family_dependencies. U12-152 says \|a x i\|^2+\|a x j\|^2+\|a x k\|^2 is not \|a\|^2 and shows in its explanation that it equals 2\|a\|^2, which answers U12-134 (2*14 = 28) at once. | **rewritten** Confirmed with sympy: sum \|a x e_k\|^2 = 28 = 2\|a\|^2, which U12-C-045 Statement II's explanation states. Rewrote U12-C-027 (Medium): given a x i = 3j + k and a x j = -3i + 2k, find a.(i + j + k) = 4 (a = 2i - j + 3k, unique). New family U12-C-recover-from-unit-cross; distractors from order reversal and unit-vector sign slips; numerical alternate 4. | U12-C-027 |
| ERR-243 | U12-B-006 JEE-MATH-U12-060 | E6 | minor | U12-065 Statement II states the key step of U12-060 (\|a+b\| = \|a-b\| means a is perpendicular to b), and the two are not linked. | **aliased** Verified: \|a+b\|^2 - \|a-b\|^2 = 4 a.b, so U12-B-011 Statement II states the key step of U12-B-006 (key 5). Different tasks, so aliased U12-B-006 with U12-B-011 rather than rewritten. | - |
| ERR-244 | U12-C-047 JEE-MATH-U12-154 | E6 | minor | U12-100 Statement I gives away the dot-product step of U12-154: with a^+b^+c^=0, the sum of the pairwise dot products is -3/2, so each one is -1/2. Not linked by dependencies. | **aliased** Related task only: U12-B-046 Statement I supplies the dot-product step (each pairwise dot = -1/2) but not the cross-product reduction or the answer 3sqrt(3)/2. Alias requested. | - |
| ERR-245 | U12-A-050 JEE-MATH-U12-050 | E6 | minor | U12-050 (octagon) is the same task as U12-022 (hexagon) with n changed: the sum of the vectors from one vertex to all the others equals n times the vector to the centre. The families differ and are not linked. | **aliased** Confirmed with sympy (hexagon sum = 6 PO, octagon sum = 8 P1O). Same principle but different polygon, different unknown form (basis vector vs scalar lambda); related task, so aliased with U12-A-022 rather than rewritten. | - |
| ERR-246 | U12-A-036 JEE-MATH-U12-036 | E6 | minor | Statement II of U12-036 (number of integer vectors of magnitude sqrt(3)) is the same counting task as U12-041 (number of integer vectors of magnitude 3) with a different number. Not linked. | **aliased** Confirmed by enumeration (8 integer vectors of magnitude sqrt(3), 30 of magnitude 3). Statement II is a small related count inside a statement item; aliased with U12-A-041. | - |

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| ERR-247 | U12-C-010 JEE-MATH-U12-117 | E6 | minor | U12-117 uses exactly the same diagonal vectors as U12-038 (3i+j-2k and i-3j+4k). U12-038's working recovers the sides, so a student who has done U12-038 can get the area in U12-117 straight from those sides. Not linked. | **aliased** Confirmed same diagonals as U12-A-038 (sides (2,-1,1), (1,2,-3), cross (1,7,5), magnitude 5sqrt(3)), but the tasks differ (longer side vs area). Related task, so alias requested rather than rewrite. | - |
| ERR-248 | U12-A-044 JEE-MATH-U12-044 | E5 | minor | The rationale for option A is not self-contained: it begins 'Same sign slip', which points to the slip described in B's rationale, listed after it. | **fixed** Rationale for the -2 distractor rewritten to be self-contained: sign slip gives lambda^2 - lambda - 6 = 0, roots 3 and -2, picks -2 (verified with sympy). | - |
| ERR-249 | U12-C-040 JEE-MATH-U12-147 | E2 | minor | 'makes an obtuse angle with the y-axis' does not say which direction of the axis is meant. A line has no direction, and with the negative y-axis the key would be the opposite vector, which is not an option, so the meaning can be inferred but is not stated. | **fixed** Stem and explanation now say 'obtuse angle with the positive y-axis'; key unchanged. | U12-C-040 |
| ERR-250 | U12-B-007 JEE-MATH-U12-061 | E9 | minor | Inconsistent notation: the explanation and distractor rationales B, D use $\vec{PQ}$-style two-letter vectors, while the rest of the unit uses $\overrightarrow{PQ}$. Also, \vec renders a short arrow over only part of the two letters. | **fixed** All two-letter \vec{XY} replaced with \overrightarrow{XY} in explanation and rationales; values re-verified (work 3, distractors -3, 7, 11). | - |
| ERR-251 | U12-B-028 JEE-MATH-U12-082 | E9 | minor | Inconsistent notation: the explanation and distractor rationales A, B, C use $\vec{PQ}$-style two-letter vectors, while the rest of the unit uses $\overrightarrow{PQ}$. Also, \vec renders a short arrow over only part of the two letters. | **fixed** All two-letter \vec{XY} replaced with \overrightarrow{XY} in explanation and rationales; cos QPR = sqrt6/3 and distractor arithmetic re-verified. | - |
| ERR-252 | U12-B-029 JEE-MATH-U12-083 | E9 | minor | Inconsistent notation: the explanation and distractor rationale C use $\vec{PQ}$-style two-letter vectors, while the rest of the unit uses $\overrightarrow{PQ}$. Also, \vec renders a short arrow over only part of the two letters. | **fixed** Two-letter \vec{XY} replaced with \overrightarrow{XY} in explanation and rationale; work 40 and distractors -40, 39, 24 re-verified. | - |
| ERR-253 | U12-B-049 JEE-MATH- | E9 | minor | Inconsistent notation: the distractor rationale D use $\vec{PQ}$-style two-letter vectors, while the rest of the unit uses $\overrightarrow{PQ}$. Also, \vec renders a short arrow over only part of the two letters. | **fixed** Two-letter \vec{PQ} replaced with \overrightarrow{PQ} in the rationale; work 22 and distractors 11, 42, 74 re-verified. | - |

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| | U12-103 | | | | | |
| ERR-254 | U13-D-023 JEE-MATH-U13-158 | E6 | major | Near-duplicate of U13-119 with trivially different numbers, not linked by family_dependencies (U13-D-bayes-reverse-draw vs U13-C-condition-on-later). Both ask: two draws without replacement from a collection of 10, given that the second draw is of type X, find P(first draw was type X). Both share the misconception text 'Believing a later draw carries no information about an earlier one.' | **rewritten** Confirmed: same task as U13-C-029 (P(1st of type \| 2nd of type) = (r-1)/(N-1), 1/3 re-verified). Rewritten as a swap urn (3 red, 5 black; drawn ball replaced by one of the other colour), given 2nd red find P(1st red) = 3/13 via total probability and Bayes; no symmetry shortcut applies. New family U13-D-bayes-swap-draw. Still Medium. | U13-D-023 |
| ERR-255 | U13-D-031 JEE-MATH-U13-166 | E6 | major | Given away by U13-125, not linked by family_dependencies. U13-125 (two cards without replacement, given at least one ace, P(both aces)) must compute P(at least one ace) = 198/1326 = 33/221 as its denominator, which is exactly the key of U13-166 (P(X >= 1) for the number of aces in the same experiment). U13-125's explanation prints this value. | **rewritten** Confirmed: key 33/221 equals the denominator printed in U13-C-035's explanation. Rewritten to number of face cards in two cards without replacement, P(X = 1) = 80/221 (brute-force check); not recoverable from U13-C-035. New family U13-D-rv-cards-face. Still Medium. | U13-D-031 |
| ERR-256 | U13-D-043 JEE-MATH-U13-178 | E6 | major | Near-duplicate of U13-042 and given away by U13-083 (and U13-025), none linked by family_dependencies. U13-042 asks for n when values 1..n with linearly varying weights have mean 7; U13-178 is the same task with the weights reversed (P(X=k) proportional to k), and U13-042's distractor C rationale is literally U13-178's solution. U13-083 (value k occurs k times, k = 1..10) computes the mean as 7 in its explanation, which is the n = 10 answer of U13-178; U13-025 is the same weighted-mean setup with n = 20. | **rewritten** Confirmed: (2n+1)/3 = 7 gives n = 10, matching U13-A-042 distractor, U13-B-038 mean 7 and U13-A-025 setup. Rewritten with P(X = k) = c/(k(k+1)) and the unknown n found from a tail probability P(X >= 2) = 9/20 (telescoping normalisation; n = 10, unique, brute-force confirmed). Distractors 19, 9, 8 from no normalisation and two off-by-one telescoping slips, each re-solved in sympy. New family U13-D-rv-pmf-telescoping-find-n. Still Difficult; numerical alternate kept (answer 10). | U13-D-043 |
| ERR-257 | U13-B-001 JEE-MATH-U13-046 | E6 | minor | U13-046 (variance of 1..10) prints the key of U13-001 (mean of 1^2..10^2 = 77/2) in both its working and its distractor A; the two items are not linked by family_dependencies. | **aliased** Confirmed related, not duplicate: sympy gives mean of 1^2..10^2 = 77/2 (U13-A-001 key) and variance of 1..10 = 77/2 - 121/4 = 33/4 (U13-B-001 key). U13-B-001 asks a different quantity (variance) with a different answer, and 77/2 is a legitimate forgot-to-subtract-mean-squared distractor, but one item exposes the other's key. Alias group requested so they never share a paper. | - |
| ERR-258 | U13-B-025 JEE-MATH-U13-070 | E6 | minor | U13-070 (statement item) explanation states that the variance of the first n even natural numbers is 4(n^2-1)/12, which is exactly the fact U13-069 tests in inverse form (and that U13-052 tests for odd numbers); not linked by family_dependencies. | **aliased** Confirmed related, not duplicate: sympy confirms variance of 2,4,...,2n is 4(n^2-1)/12 (U13-B-025 Statement II, false as 'twice'), m = 5 from 4(m^2-1)/12 = 8 in U13-B-024, and variance of 1,3,...,19 = 33 in U13-B-007. Each asks a different task (statement truth, inverse m+n = 24, direct value 33), so no rewrite; 025's explanation states the scaling fact the others use. Alias group requested. | - |
| ERR-259 | U13-B-005 JEE-MATH-U13-050 | E6 | minor | The mean part of U13-050 is the same task as U13-003 (mean of a*x + b given the mean) with trivially different numbers; the items are in different families (U13-B-origin-scale vs U13-A-mean-linear-transform) with no shared dependency. | **aliased** Confirmed related, not duplicate: U13-B-005 mean 3(10)+2 = 32 repeats the mean rule of U13-A-003 (3(15)-4 = 41) with different numbers, but the discriminating part of 005 is the variance 9*4 = 36 (all options share the mean), which A-003 does not test. Different data and answer, so no rewrite. Alias group requested. | - |

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| ERR-260 | U13-A-035 JEE-MATH-U13-035 | E6 | minor | Statement II of U13-035 (median of 3x - 2 is 3M - 2) is the general form of U13-013 (median of -2x + 5 given median 12); not linked by family_dependencies, so one gives away the principle of the other in the same paper. | **aliased** Confirmed with sympy: Statement II of U13-A-035 (median of 3x-2 is 3M-2) states the general rule that U13-A-013 applies (-2(12)+5 = -19). Related principle, different data and answer, so not rewritten; alias requested so they never share a paper. Records carry no family_dependencies field, so the link goes through alias_requests. | - |
| ERR-261 | U13-A-040 JEE-MATH-U13-040 | E6 | minor | U13-040 (block overlap of partial means) and U13-029 (single shared observation in overlapping partial means) are the same inclusion-exclusion-on-sums task with different numbers, in different families without a shared dependency. | **aliased** Confirmed with sympy: both use inclusion-exclusion on partial sums (52+64=105+x4 gives 11; 2880+3300=5000+S gives S=1180, mean 59). Different data, unknown (single term vs block mean) and difficulty, so related rather than duplicate; alias requested. | - |
| ERR-262 | U13-A-019 JEE-MATH-U13-019 | E6 | minor | U13-019 (missing frequency of grouped data from the mean) is the same task as U13-009 (missing frequency of a discrete distribution from the mean) with class marks in place of values; families differ and no dependency is shared. | **aliased** Confirmed with sympy: same missing-frequency-from-mean equation (k=5 discrete, k=12 grouped with class marks). Different data and answers, and the grouped item adds a class-mark step, so related rather than duplicate; alias requested. | - |
| ERR-263 | U13-B-023 JEE-MATH-U13-068 | E6 | minor | U13-068 (two unknown observations a, b from mean and variance; find ab) is the same reconstruction task as U13-051 (two observations from mean and SD; find their product), in different families without a shared dependency. | **aliased** Confirmed related, not duplicate: sympy solves U13-B-006 to {4, 10}, ab = 40, and U13-B-023 to {6, 8}, ab = 48 (a+b = 14, a^2+b^2 = 210-110 = 100). Same reconstruction method but different data, different answers, and 023 adds removal of three known observations, so no rewrite. Alias group requested. | - |
| ERR-264 | U13-B-037 JEE-MATH-U13-082 | E6 | minor | U13-082 (least value of the mean deviation about a point, attained at the median) proves the property that is Statement II of U13-055, and U13-075's explanation also restates it; the three items are not linked. | **aliased** Confirmed related, not duplicate: sympy gives least of (1/6) sum \|x_i - a\| = 19/3 for U13-B-037; U13-B-030 ratio 17/18 for 2,6,7,9,16; U13-B-010 Statement II (median minimises mean deviation) is the property 037 applies and 030's explanation cites. Different tasks and answers, so no rewrite. Alias group requested. | - |
| ERR-265 | U13-B-015 JEE-MATH-U13-060 | E9 | minor | The stem ends 'the mean deviation about the median is:' and then gives the data after the colon, so the data reads as if it were the answer. Other items in the slice give the data first, or put it on its own line (e.g. U13-002). | **fixed** Stem of U13-B-015 reworded so the data precedes the question: 'The values $x$: $3, 5, 7, 9, 11$ occur with frequencies $f$: $2, 4, 6, 5, 3$ respectively. The mean deviation about the median is'. Key re-verified with sympy: median 7, sum f\|x-7\| = 38, MD = 19/10; distractor 49/25 re-verified (mean 7.3, sum 39.2). | U13-B-015 |

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| ERR-266 | U13-C-002 JEE-MATH-U13-092 | E9 | minor | Numerals in the stem are written outside $...$, which is inconsistent with the rest of Unit 13 (batches U13-A, B and D put every number in math mode) and with the bank's latex-inline notation. | **fixed** Verified bare numerals in stem; wrapped every stem number in $...$ (also in the numerical alternate stem and in explanation/rationale/difficulty_reason text where present). No change to maths, key or options. | U13-C-002 |
| ERR-267 | U13-C-004 JEE-MATH-U13-094 | E9 | minor | Numerals in the stem are written outside $...$, which is inconsistent with the rest of Unit 13 (batches U13-A, B and D put every number in math mode) and with the bank's latex-inline notation. | **fixed** Verified bare numerals in stem; wrapped every stem number in $...$ (also in the numerical alternate stem and in explanation/rationale/difficulty_reason text where present). No change to maths, key or options. | U13-C-004 |
| ERR-268 | U13-C-006 JEE-MATH-U13-096 | E9 | minor | Numerals in the stem are written outside $...$, which is inconsistent with the rest of Unit 13 (batches U13-A, B and D put every number in math mode) and with the bank's latex-inline notation. | **fixed** Verified bare numerals in stem; wrapped every stem number in $...$ (also in the numerical alternate stem and in explanation/rationale/difficulty_reason text where present). No change to maths, key or options. | U13-C-006 |
| ERR-269 | U13-C-007 JEE-MATH-U13-097 | E9 | minor | Numerals in the stem are written outside $...$, which is inconsistent with the rest of Unit 13 (batches U13-A, B and D put every number in math mode) and with the bank's latex-inline notation. | **fixed** Verified bare numerals in stem; wrapped every stem number in $...$ (also in the numerical alternate stem and in explanation/rationale/difficulty_reason text where present). No change to maths, key or options. | U13-C-007 |
| ERR-270 | U13-C-009 JEE-MATH-U13-099 | E9 | minor | Numerals in the stem are written outside $...$, which is inconsistent with the rest of Unit 13 (batches U13-A, B and D put every number in math mode) and with the bank's latex-inline notation. | **fixed** Verified bare numerals in stem; wrapped every stem number in $...$ (also in the numerical alternate stem and in explanation/rationale/difficulty_reason text where present). No change to maths, key or options. | U13-C-009 |
| ERR-271 | U13-C-012 JEE-MATH-U13-102 | E9 | minor | Numerals in the stem are written outside $...$, which is inconsistent with the rest of Unit 13 (batches U13-A, B and D put every number in math mode) and with the bank's latex-inline notation. | **fixed** Verified bare numerals in stem; wrapped every stem number in $...$ (also in the numerical alternate stem and in explanation/rationale/difficulty_reason text where present). No change to maths, key or options. | U13-C-012 |

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| ERR-272 | U13-C-013 JEE-MATH-U13-103 | E9 | minor | Numerals in the stem are written outside $...$, which is inconsistent with the rest of Unit 13 (batches U13-A, B and D put every number in math mode) and with the bank's latex-inline notation. | **fixed** Verified bare numerals in stem; wrapped every stem number in $...$ (also in the numerical alternate stem and in explanation/rationale/difficulty_reason text where present). No change to maths, key or options. | U13-C-013 |
| ERR-273 | U13-C-017 JEE-MATH-U13-107 | E9 | minor | Numerals in the stem are written outside $...$, which is inconsistent with the rest of Unit 13 (batches U13-A, B and D put every number in math mode) and with the bank's latex-inline notation. | **fixed** Verified bare numerals in stem; wrapped every stem number in $...$ (also in the numerical alternate stem and in explanation/rationale/difficulty_reason text where present). No change to maths, key or options. | U13-C-017 |
| ERR-274 | U13-C-019 JEE-MATH-U13-109 | E9 | minor | Numerals in the stem are written outside $...$, which is inconsistent with the rest of Unit 13 (batches U13-A, B and D put every number in math mode) and with the bank's latex-inline notation. | **fixed** Verified bare numerals in stem; wrapped every stem number in $...$ (also in the numerical alternate stem and in explanation/rationale/difficulty_reason text where present). No change to maths, key or options. | U13-C-019 |
| ERR-275 | U13-C-022 JEE-MATH-U13-112 | E9 | minor | Numerals in the stem are written outside $...$, which is inconsistent with the rest of Unit 13 (batches U13-A, B and D put every number in math mode) and with the bank's latex-inline notation. | **fixed** Verified bare numerals in stem; wrapped every stem number in $...$ (also in the numerical alternate stem and in explanation/rationale/difficulty_reason text where present). No change to maths, key or options. | U13-C-022 |
| ERR-276 | U13-C-024 JEE-MATH-U13-114 | E9 | minor | Numerals in the stem are written outside $...$, which is inconsistent with the rest of Unit 13 (batches U13-A, B and D put every number in math mode) and with the bank's latex-inline notation. | **fixed** Verified bare numerals in stem; wrapped every stem number in $...$ (also in the numerical alternate stem and in explanation/rationale/difficulty_reason text where present). No change to maths, key or options. | U13-C-024 |
| ERR-277 | U13-C-029 JEE-MATH-U13-119 | E9 | minor | Numerals in the stem are written outside $...$, which is inconsistent with the rest of Unit 13 (batches U13-A, B and D put every number in math mode) and with the bank's latex-inline notation. | **fixed** Verified bare numerals in stem; wrapped every stem number in $...$ (also in the numerical alternate stem and in explanation/rationale/difficulty_reason text where present). No change to maths, key or options. | U13-C-029 |

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| ERR-278 | U13-C-031 JEE-MATH-U13-121 | E9 | minor | Numerals in the stem are written outside $...$, which is inconsistent with the rest of Unit 13 (batches U13-A, B and D put every number in math mode) and with the bank's latex-inline notation. | **fixed** Verified bare numerals in stem; wrapped every stem number in $...$ (also in the numerical alternate stem and in explanation/rationale/difficulty_reason text where present). No change to maths, key or options. | U13-C-031 |
| ERR-279 | U13-C-033 JEE-MATH-U13-123 | E9 | minor | Numerals in the stem are written outside $...$, which is inconsistent with the rest of Unit 13 (batches U13-A, B and D put every number in math mode) and with the bank's latex-inline notation. | **fixed** Verified bare numerals in stem; wrapped every stem number in $...$ (also in the numerical alternate stem and in explanation/rationale/difficulty_reason text where present). No change to maths, key or options. | U13-C-033 |
| ERR-280 | U13-C-035 JEE-MATH-U13-125 | E9 | minor | Numerals in the stem are written outside $...$, which is inconsistent with the rest of Unit 13 (batches U13-A, B and D put every number in math mode) and with the bank's latex-inline notation. | **fixed** Verified bare numerals in stem; wrapped every stem number in $...$ (also in the numerical alternate stem and in explanation/rationale/difficulty_reason text where present). No change to maths, key or options. | U13-C-035 |
| ERR-281 | U13-C-036 JEE-MATH-U13-126 | E9 | minor | Numerals in the stem are written outside $...$, which is inconsistent with the rest of Unit 13 (batches U13-A, B and D put every number in math mode) and with the bank's latex-inline notation. | **fixed** Verified bare numerals in stem; wrapped every stem number in $...$ (also in the numerical alternate stem and in explanation/rationale/difficulty_reason text where present). No change to maths, key or options. | U13-C-036 |
| ERR-282 | U13-C-037 JEE-MATH-U13-127 | E9 | minor | Numerals in the stem are written outside $...$, which is inconsistent with the rest of Unit 13 (batches U13-A, B and D put every number in math mode) and with the bank's latex-inline notation. | **fixed** Verified bare numerals in stem; wrapped every stem number in $...$ (also in the numerical alternate stem and in explanation/rationale/difficulty_reason text where present). No change to maths, key or options. | U13-C-037 |
| ERR-283 | U13-C-042 JEE-MATH-U13-132 | E9 | minor | Numerals in the stem are written outside $...$, which is inconsistent with the rest of Unit 13 (batches U13-A, B and D put every number in math mode) and with the bank's latex-inline notation. | **fixed** Verified bare numerals in stem; wrapped every stem number in $...$ (also in the numerical alternate stem and in explanation/rationale/difficulty_reason text where present). No change to maths, key or options. | U13-C-042 |

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| ERR-284 | U13-C-043 JEE-MATH-U13-133 | E9 | minor | Numerals in the stem are written outside $...$, which is inconsistent with the rest of Unit 13 (batches U13-A, B and D put every number in math mode) and with the bank's latex-inline notation. | **fixed** Verified bare numerals in stem; wrapped every stem number in $...$ (also in the numerical alternate stem and in explanation/rationale/difficulty_reason text where present). No change to maths, key or options. | U13-C-043 |
| ERR-285 | U13-C-044 JEE-MATH-U13-134 | E9 | minor | Numerals in the stem are written outside $...$, which is inconsistent with the rest of Unit 13 (batches U13-A, B and D put every number in math mode) and with the bank's latex-inline notation. | **fixed** Verified bare numerals in stem; wrapped every stem number in $...$ (also in the numerical alternate stem and in explanation/rationale/difficulty_reason text where present). No change to maths, key or options. | U13-C-044 |
| ERR-286 | U14-C-034 JEE-MATH-U14-144 | E6 | major | Near-duplicate of U14-014 with identical numbers and answer, and its answer is given away by U14-126; no shared family_dependencies with either. | **rewritten** Confirmed: tan^-1 2 + tan^-1 3 + tan^-1 k = pi with k = 1 is the same task, numbers and key as U14-A-014, and U14-C-016 (sum = 3pi/4) gives it away. Rewritten as a different unknown and twist: tan^-1 k + tan^-1(2k) + tan^-1(3k) = pi/2, needing a sign argument for k > 0 and the pi/2 condition; sympy gives the unique root k = 1/sqrt(11). New distractors from real slips (pi condition gives 1; 1 + xy denominator gives 1/sqrt(7); dropped pairwise product gives 1/sqrt(5)), new explanation, numerical alternate now asks 1/k^2 = 11, new family U14-C-tan-three-sum-param. Still Medium. | U14-C-034 |
| ERR-287 | U14-C-037 JEE-MATH-U14-147 | E6 | minor | The key sub-result (tan^-1 2 + tan^-1 3 = 3pi/4) is exactly the answer of U14-126, and the families are not linked. | **aliased** Confirmed related, not duplicate: 147 (cot^-1(-1/2) + cot^-1(-1/3) = 5pi/4, sympy confirms) uses tan^-1 2 + tan^-1 3 = 3pi/4 from U14-C-016 as an intermediate step but adds the negative-argument rule and a different answer. Alias U14-C-037 with U14-C-016 requested. | - |
| ERR-288 | U14-C-023 JEE-MATH-U14-133 | E6 | minor | Statement I is the identity that U14-018 asks for, so a paper containing both gives one away; families are not linked. 2tan^-1(1/3) = tan^-1(3/4) is also the main step of U14-135. | **rewritten** Confirmed: Statement I (2tan^-1(1/3) + tan^-1(1/7) = pi/4) states the answer of U14-A-018 directly and repeats the main step of U14-C-025. Statement I replaced by tan^-1(1/2) + tan^-1(1/5) + tan^-1(1/8) = pi/4 (sympy confirms true), so the key (both true) is unchanged; Statement II unchanged. Rationales for the Statement I false options rewritten self-contained (dropping 1 - xy gives tan^-1(66/73), sympy confirms), explanation and check updated. | U14-C-023 |
| ERR-289 | U14-C-044 JEE-MATH-U14-154 | E6 | minor | Same telescoping task as U14-130 (tan^-1 of 1/(1+m(m+1)) split as tan^-1(m+1) - tan^-1 m, then tan of the sum), with an index shift and the reverse direction; families are not linked. | **aliased** Confirmed related, not duplicate: 154 uses the same telescoping split as 130 but with cot^-1 input, an index shift and an inverse unknown (n = 10, sympy confirms). Alias U14-C-044 with U14-C-020 requested. | - |

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| ERR-290 | U14-A-023 JEE-MATH-U14-023 | E6 | minor | Same doubling-telescoping cosine product as U14-005 and U14-024 (which are linked to each other), but 023 is not linked to them. | **aliased** Confirmed related, not duplicate: sympy confirms cos(pi/15)cos(2pi/15)cos(4pi/15)cos(8pi/15) = -1/16 via the same doubling telescoping as 005 (cos20cos40cos80 = 1/8) and 024 (six factors, k = 6). 023 has different data, a different answer and an extra allied-angle sign twist (sin(16pi/15) = -sin(pi/15)), so no rewrite. The bank has no family_dependencies field, so an alias group 023/005/024 is requested so the three never share a paper. | - |
| ERR-291 | U14-B-042 JEE-MATH-U14-097 | E8 | minor | The item is effectively a trigonometric equation solved in an interval (the explanation solves tan x = +-1 and lists x = pi/4, 3pi/4 + 2k pi). The U14 scope brief lists 'Trigonometric equations and general solutions' as out of scope; only the U14-B batch prompt allows 'number of solutions only via graphs'. The brief and the batch prompt do not agree. | **rewritten** Ruling applied: trigonometric equations solved in an interval are out of scope, so U14-B-042 (count of \|cos x\| = sin x on [0, 5pi], key 6, which sympy grid count confirms) was replaced in place, same difficulty (Medium) and class (XI). New item: number of integers in the range of 8(sin^6 x + cos^6 x); sympy gives range [2, 8], key 7; distractors 3 (cube identity without factor 3, range [6, 8]), 9 (minimum taken as 0), 13 (sin^2 x cos^2 x = (1/2) sin^2 2x, range [-4, 8]) all re-derived. New family U14-B-sixth-power-range; numerical alternate kept (answer 7). | U14-B-042 |
| ERR-292 | U14-B-014 JEE-MATH-U14-069 | E8 | minor | Counting solutions of \|cos x\| = 1/2 in [0, 3pi] is solving a trigonometric equation in an interval, which conflicts with the U14 scope brief (trig equations out). Same brief/batch-prompt conflict as 097. | **rewritten** Ruling applied: counting solutions of \|cos x\| = 1/2 on [0, 3pi] is solving a trigonometric equation in an interval, so U14-B-014 was replaced in place, same difficulty (Easy) and class (XI). New item: range of 7 - 3\|cos 2x\| is [m, M], find m + M; sympy gives [4, 7], key 11; distractors 14 (modulus dropped, [4, 10]), 8 (amplitude doubled by the 2x, [1, 7]), 17 (sign slip, 7 + 3\|cos 2x\|, [7, 10]). New family U14-B-abs-shift-range; numerical alternate kept (answer 11). | U14-B-014 |
| ERR-293 | U14-B-044 JEE-MATH-U14-099 | E7 | minor | question_type is 'Inverse/parameter MCQ', but the item has no parameter; it counts the x in [0, 2pi] where 2 sin x + sqrt5 cos x reaches its maximum. | **fixed** Confirmed: no parameter in U14-B-044. question_type changed to Conceptual/application MCQ. Key re-verified with sympy: solveset gives the single value atan(2/sqrt5) in [0, 2pi], count 1. | - |
| ERR-294 | U14-B-054 JEE-MATH-U14-109 | E7 | minor | question_type is 'Graph/diagram MCQ', but the intended solution is a bound (AM-GM) argument with no graph. | **fixed** Confirmed: U14-B-054 is solved by a bound (AM-GM) argument, not a graph. question_type changed to Multi-step computation MCQ. Key 0 re-verified: at x = 1 the left side is 2cos^2(1/2)sin^2(1) = 1.0906 < 2. | - |
| ERR-295 | U14-B-007 JEE-MATH-U14-062 | E5 | minor | The rationale for distractor C contradicts itself: if both factors were negative their product would be positive, so this reading cannot follow from sin(theta)cos(theta) < 0. | **fixed** Confirmed: the old rationale was self-contradictory. Rewritten for the third-quadrant distractor of U14-B-007: keeps only the case cos theta < 0 from sin theta cos theta < 0, then a sign slip in cos theta tan theta > 0 gives tan theta > 0 instead of < 0; cos < 0 with tan > 0 is the third quadrant (sign table checked). Option text unchanged. | - |

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| ERR-296 | U14-B-016 JEE-MATH-U14-071 | E5 | minor | The rationale for distractor C says the student 'adds' the separate maxima in both statements, but Statement II is about a product; the slip there is treating sin x = cos x = 1 together (1 x 1). | **fixed** Confirmed: Statement II is about a product. Rationale for the both-true distractor of U14-B-016 rewritten: takes sin x and cos x equal to 1 (and -1) at the same x in both statements, giving range [-2, 2] and product maximum 1 x 1 = 1. Option text unchanged. | - |
| ERR-297 | U14-A-037 JEE-MATH-U14-037 | E5 | minor | The rationale for distractor C refers to 'the same mix-up with the cosine triple-angle formula', but that mix-up is only described later, in dr D. Read alone, the rationale is unclear. | **fixed** Confirmed: the rationale for the both-false distractor referred back to a mix-up only described in another rationale. Rewritten self-contained: evaluates Statement I correctly as -3/8 (sympy confirms) but rejects Statement II, believing sin^3 theta = (sin 3theta + 3 sin theta)/4 by analogy with cos 3theta = 4cos^3 theta - 3cos theta. Option text unchanged. | - |
| ERR-298 | U14-C-015 JEE-MATH-U14-125 | E5 | minor | The rationale for distractor A says 'Same double counting', which points forward to dr D; it is not self-contained. | **fixed** Confirmed: rationale for 2 - sqrt(3) pointed to another rationale. Rewritten self-contained: keeps 2cot^-1 x after using tan^-1 x + cot^-1 x = pi/2, so cot^-1 x = pi/12, then writes x = tan(pi/12) = 2 - sqrt(3). Option text unchanged. | - |
| ERR-299 | U14-C-022 JEE-MATH-U14-132 | E5 | minor | The rationale for distractor B says 'Same sign slip', which points forward to dr C; it is not self-contained. | **fixed** Confirmed: rationale for {1} pointed to another rationale. Rewritten self-contained: sign slip gives 6x^2 - 5x - 1 = 0 (roots 1, -1/6, sympy confirms), rejects the negative root and keeps x = 1. Option text unchanged. | - |
| ERR-300 | U02-C-042 JEE-MATH-U02-152 | E2 | critical | Re-verification: 'real and less than 5 (possibly equal)' can be read as 'less than or equal to 5'; under that reading k = 5 also qualifies and no option is correct. | **fixed** Stem now reads '(not necessarily distinct) are real and strictly less than 5'. Key unchanged. Re-verified independently (V101-7). | U02-C-042 |
| ERR-301 | U02-C-050 JEE-MATH-U02-160 | E2 | minor | Re-verification: 'lie in the open interval (1, 5) (possibly equal)' is ambiguous about what may be equal. | **fixed** Stem now reads 'both roots ... (not necessarily distinct) are real and lie in the open interval (1, 5)'. Key unchanged. Re-verified (V101-7). | U02-C-050 |
| ERR-302 | U02-C-031 JEE-MATH- | E2 | minor | Editor sweep: the same '(possibly equal)' phrase appears in this stem. | **fixed** Reworded to '(not necessarily distinct)'. Key unchanged. Re-verified (V101-7). | U02-C-031 |

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| | U02-141 | | | | | |
| ERR-303 | U02-D-051 JEE-MATH-U02-216 | E6 | minor | Post-fix duplicate scan: the rewritten item uses the same recurrence on x^2 - 5x + c as U02-D-025 and has the same answer 5. | **aliased**<br>Alias group ALIAS-ERR-DUPSCAN-01 links U02-D-025 and U02-D-051. | - |
| ERR-304 | U14-B-041 JEE-MATH-U14-096 | E8 | minor | Fixer query: items counting solutions of a trigonometric curve against a non-trigonometric curve (U14-B-040, 041, 043, 045, 046) and range-attainment counts (U14-B-044, 054) might fall under the ruling that trigonometric equations are out of scope. | **retained**<br>Editor ruling: solving trigonometric equations by general solution is out of scope and was removed (U14-B-014, U14-B-042). Counting intersections from graphs of trigonometric functions, and deciding whether a value lies in the range, test the graphs and ranges of trigonometric functions, which are in the syllabus. These items are retained. | - |

All review was done by AI model agents. No external human subject-expert review is claimed.